Microbiome Data and Analysis

BIOM 643 Bioinformatics

2026-03-31

Anastasia Lucas, University of Pennsylvania

Objectives

- Define the microbiome
- Understand the microbiome’s contributions to human health and disease
- Know the two main experimental techniques for studying the microbiome
- Describe traditional microbiome metrics and analysis methods
  - + considerations for microbiome analysis
  - + review microbiome data resources
  - + new paradigm & function-first approaches

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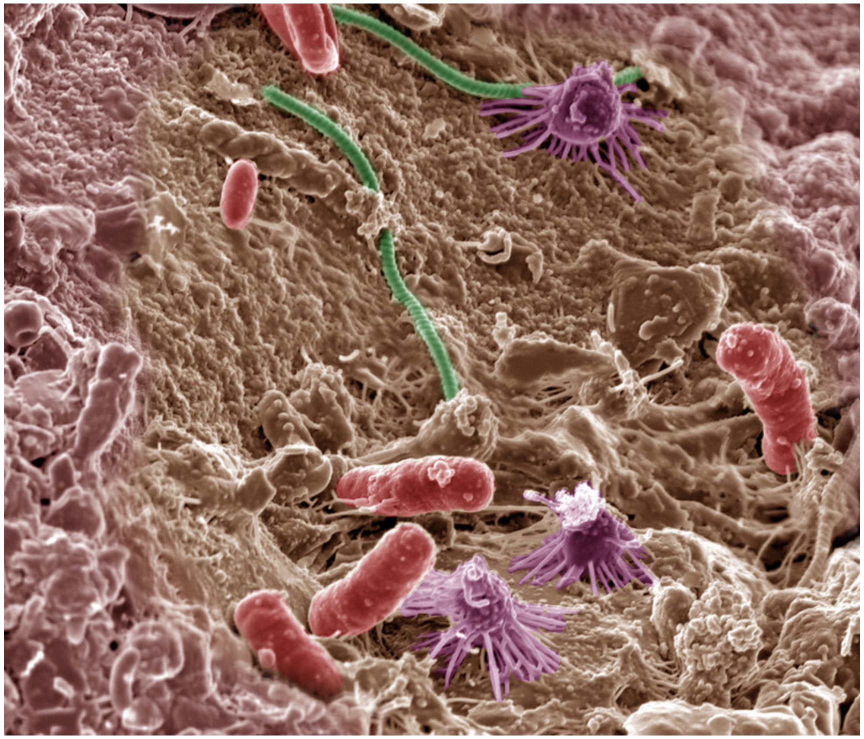
What are microbes?

Living organisms that are too small to be seen with the naked eye

Types of microbes: Bacteria, Archaea, Fungi, Protists, Viruses, Microscopic Animals

People who study the microbiome mostly talk about **bacteria**, viruses, and bacteriophages (virus that infects bacteria)



What is the microbiome?

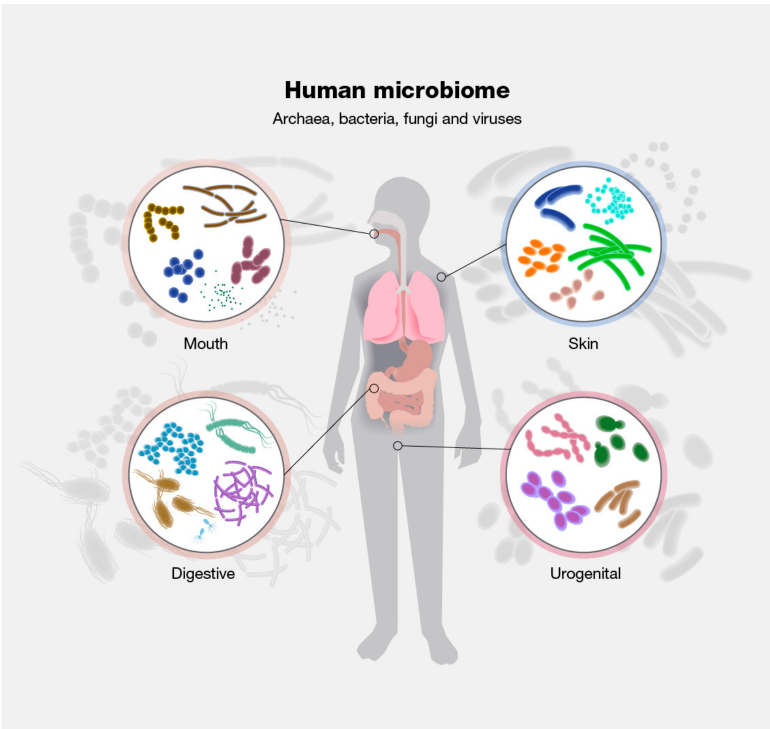


Image: NHGRI; Hartman & Six, 2022; Tara Foundation et al, 2022

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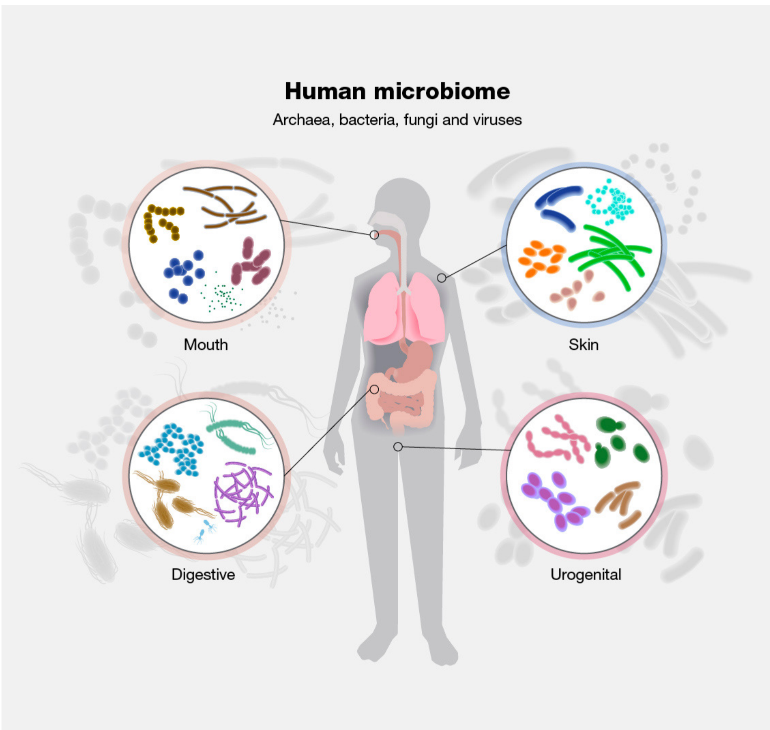

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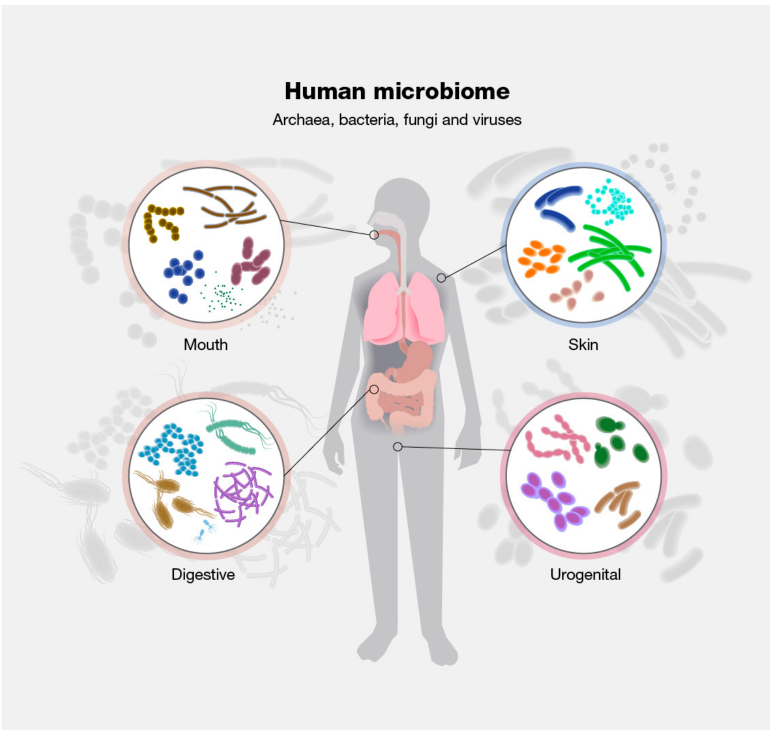

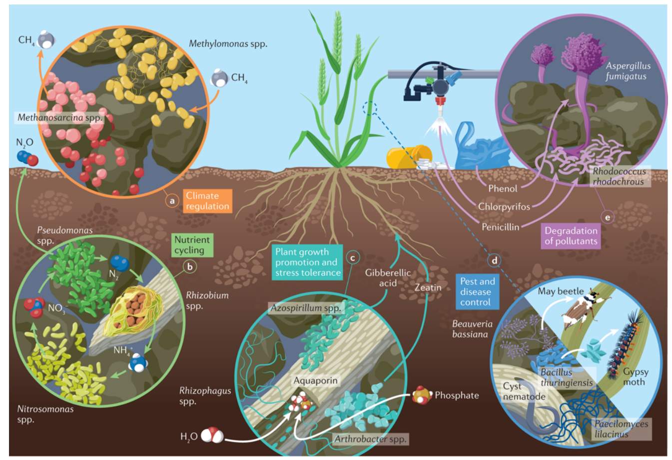

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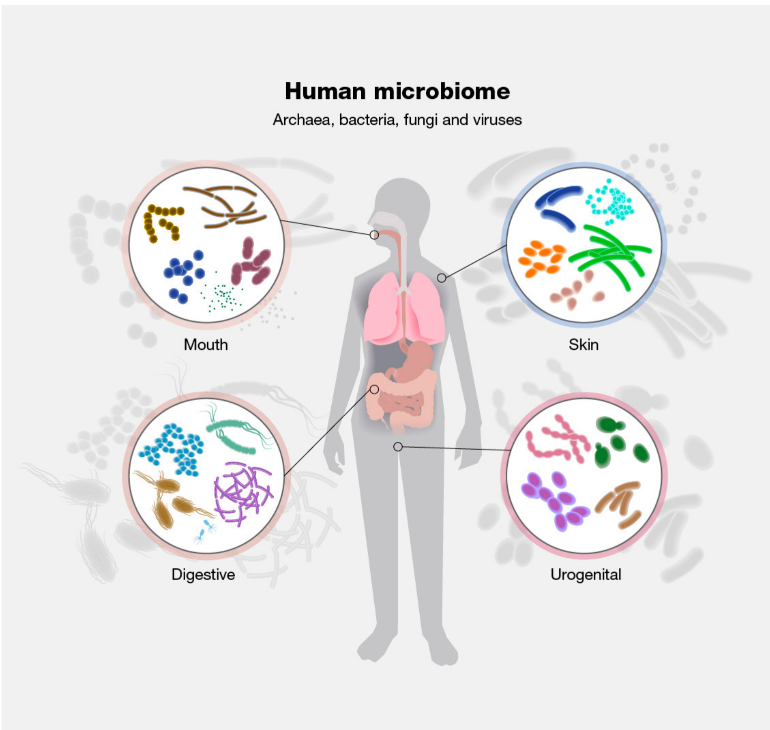

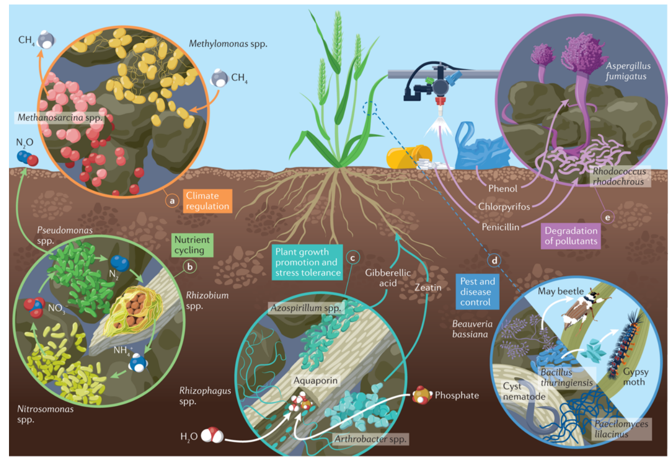

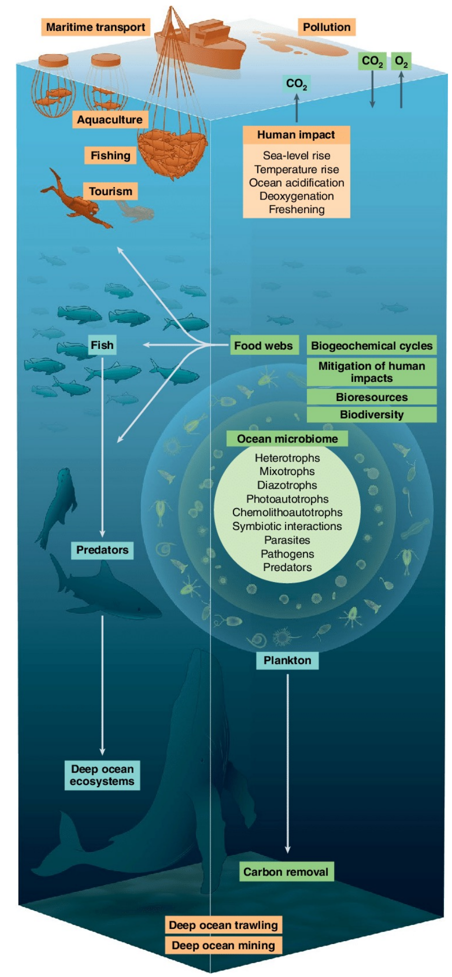

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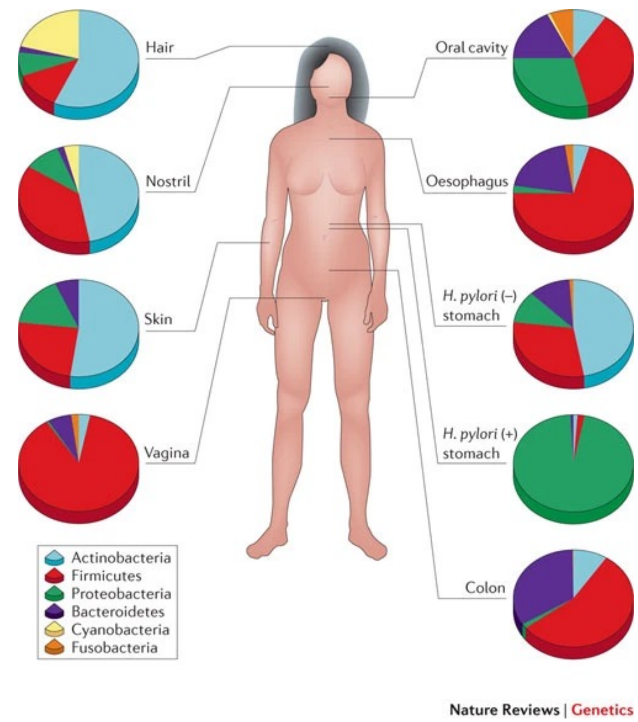
The Human Microbiome

Historical estimate
Ratio of bacterial cells to human cells in the body is 10:1

Revised estimate
39 trillion bacterial cells to 30 trillion human cells
2,000,000 genes to 20,000 genes

Humans & microbes have co-evolved, leading to cooperative interactions in metabolism, immune function, and homeostasis

Location	Typical concentration of bacteria [1] (number/mL content)
Colon (large intestine)	10^{11}
Dental plaque	10^{11}
Ileum (lower small intestine)	10^{8}
Saliva	10^{9}
Skin	$<10^{11}$ per m^2 [3]
Stomach	10^{3}–10^{4}
Duodenum and Jejunum (upper small intestine)	10^{3}–10^{4}



Images: Sender, Fuchs, & Milo, 2016; Cho & Blaser, 2012

Functions of the human microbiome

Digestion
Fermentation of dietary fiber into short-chain fatty acids

Metabolism

May synthesize vitamins

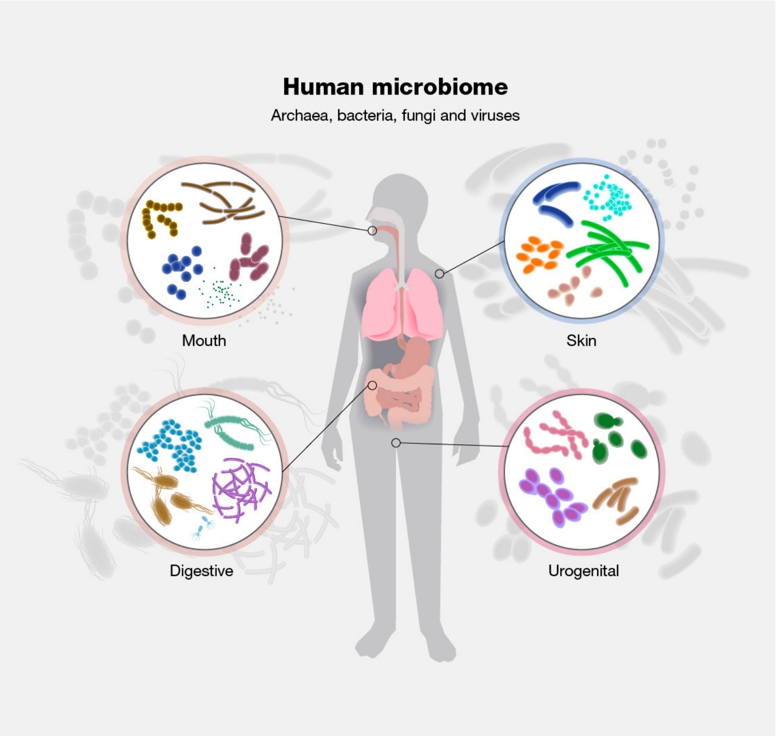


Image: NHGRI

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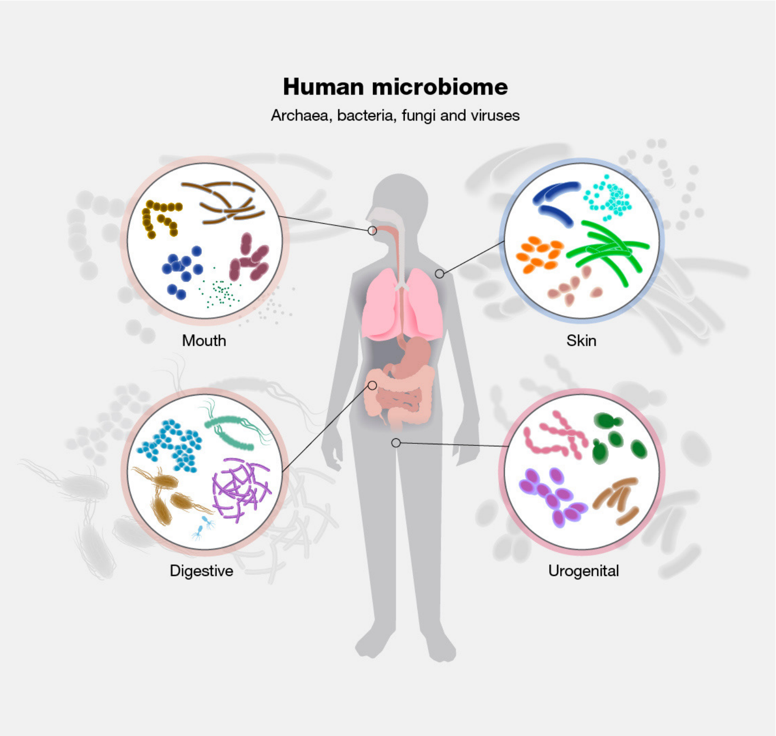
Colonization resistance
Prevents infections suppresses pathogens in the oral cavity, reproductive system, lungs



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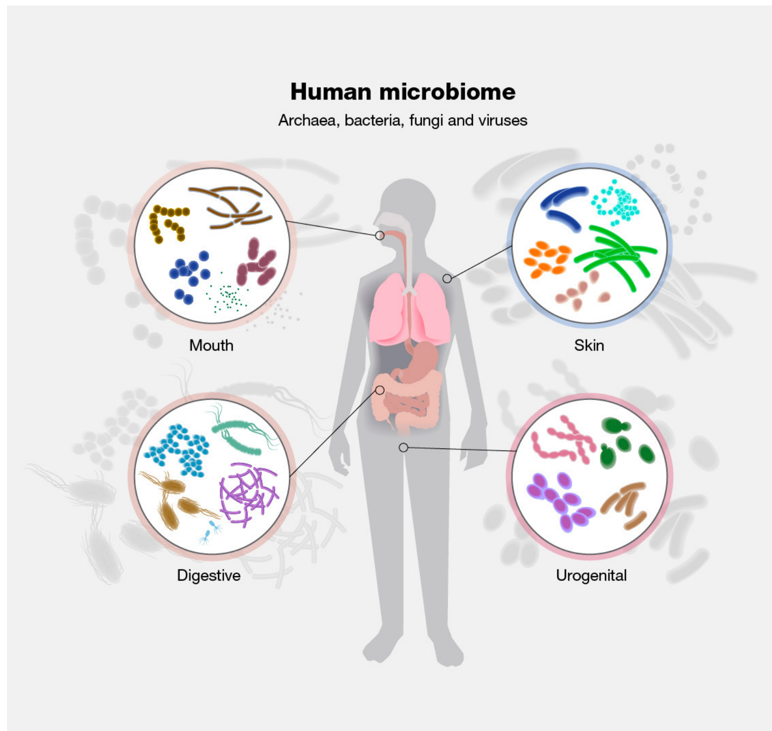

Behavior
The gut-microbiota-brain axis allows bidirectional communication between gut microbes and the brain, e.g. hunger

Image: NHGRI

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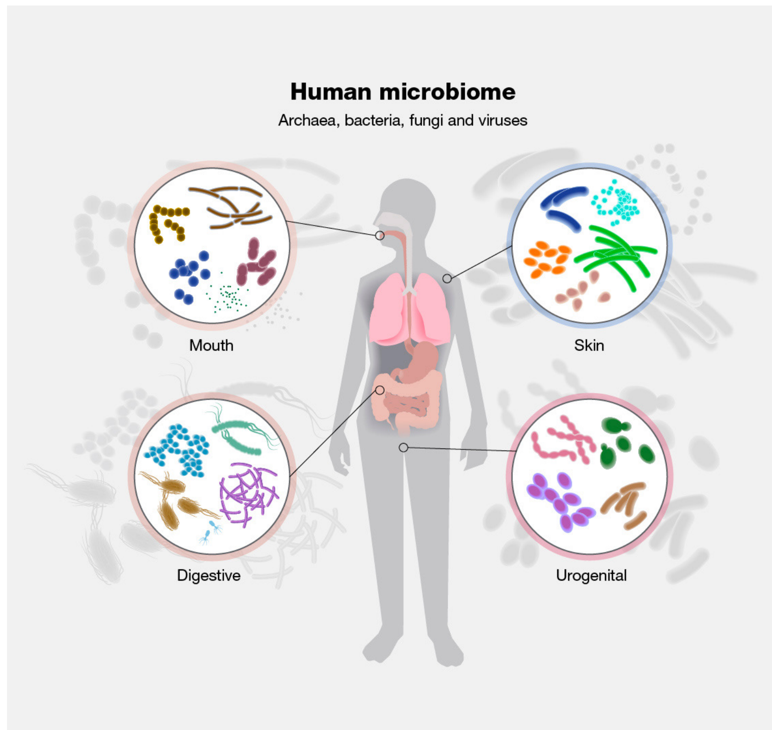

Behavior
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Immune regulation & homeostasis
Early life colonization

Immune system needs to tolerate helpful microbes

Dysregulation can lead to inflammatory bowel disease, celiac, rheumatic arthritis, etc.

Image: NHGRI

Many factors influence the human microbiome over the course of life

Infant microbiome is more heavily influenced by the mother

Microbiome is more stable after childhood

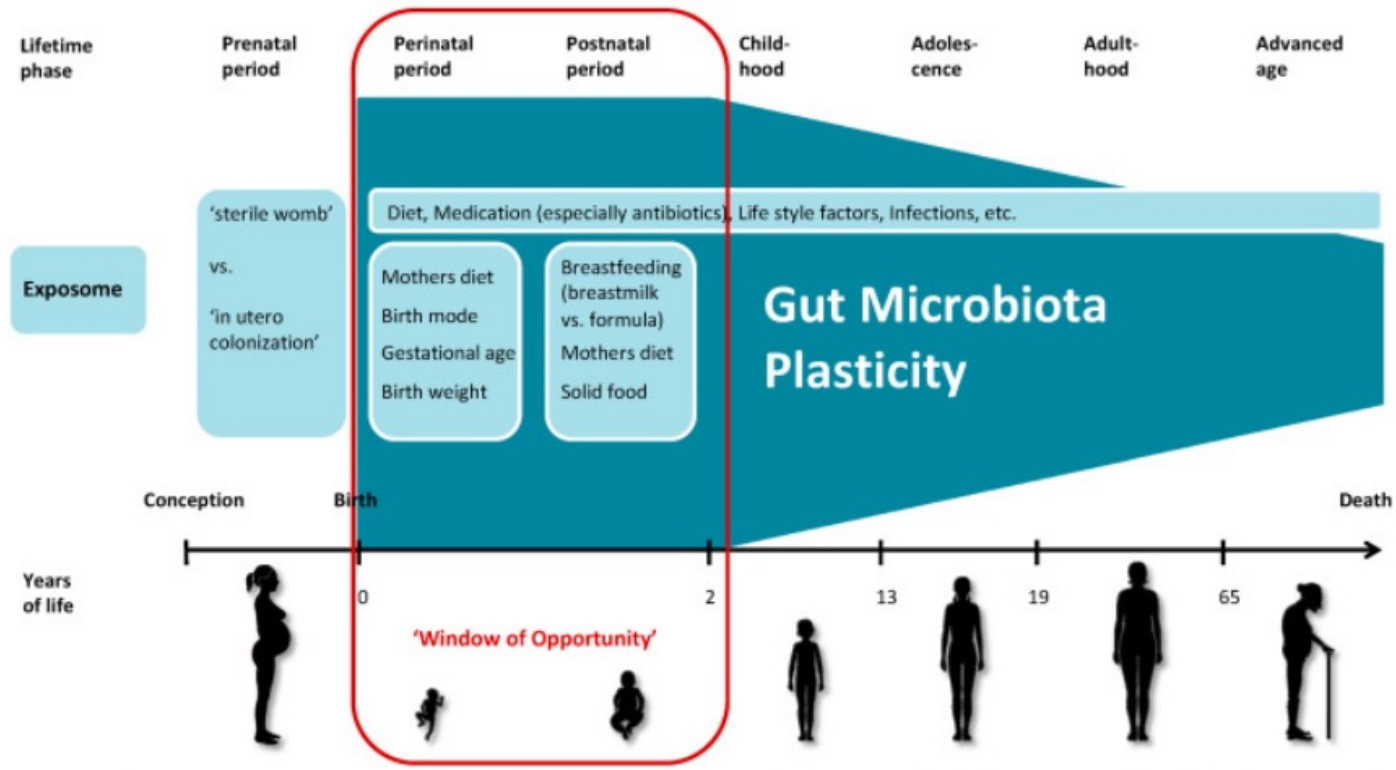

Image: Thriene & Michels, 2023

Environmental factors impact the microbiome

Cohabitation is associated with lower distances between microbiomes

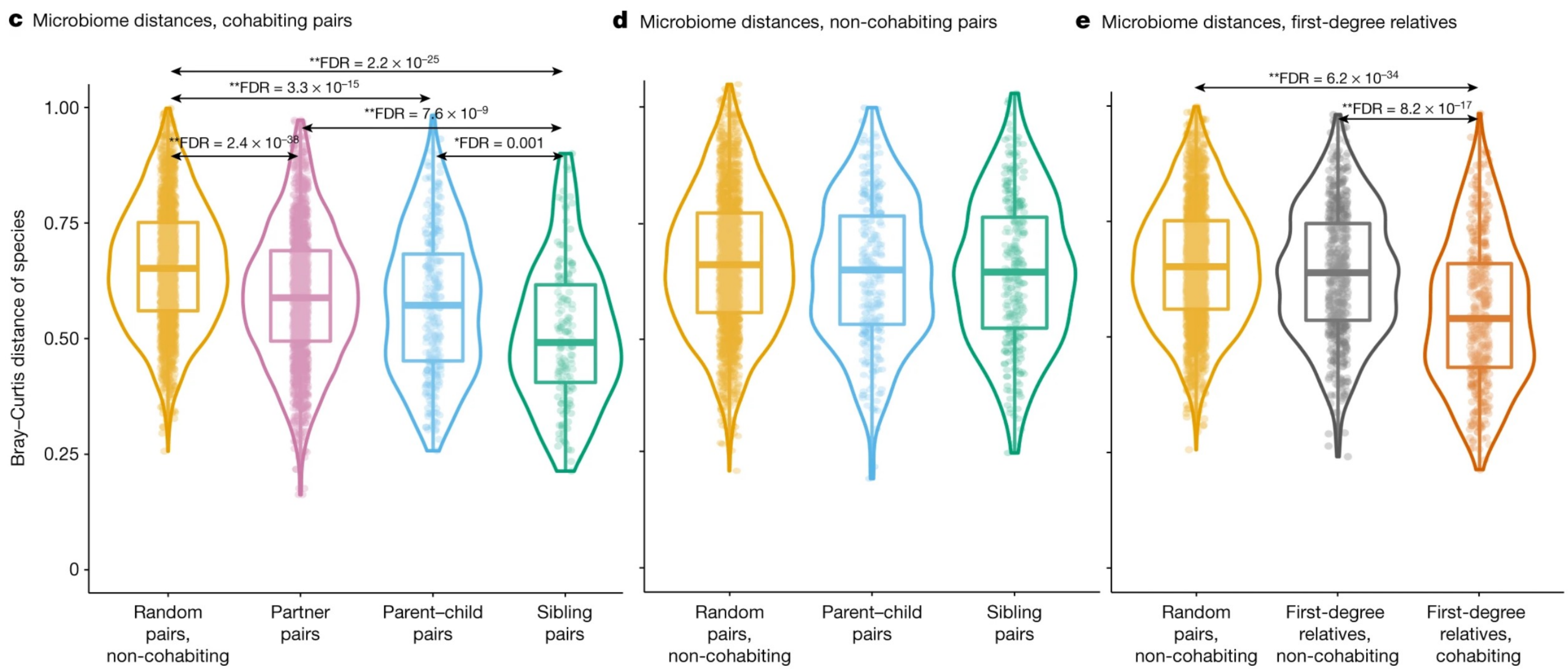

Image: Gacesa et al, 2022

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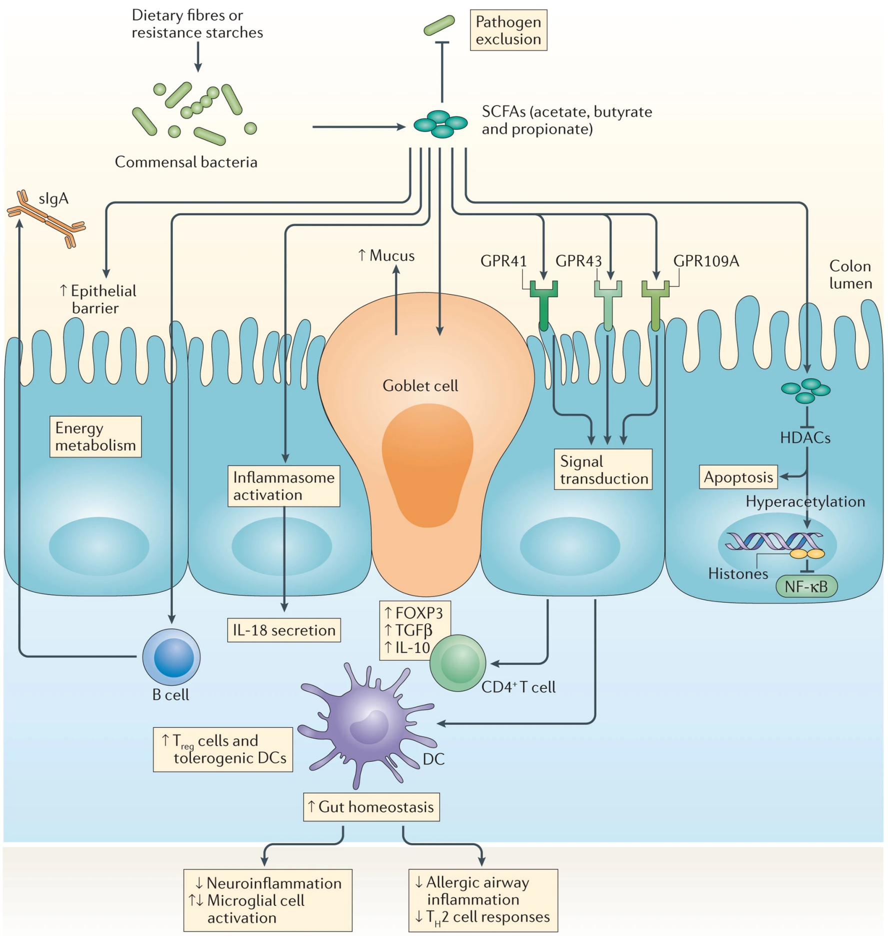
The gut microbiome and the immune system are intertwined



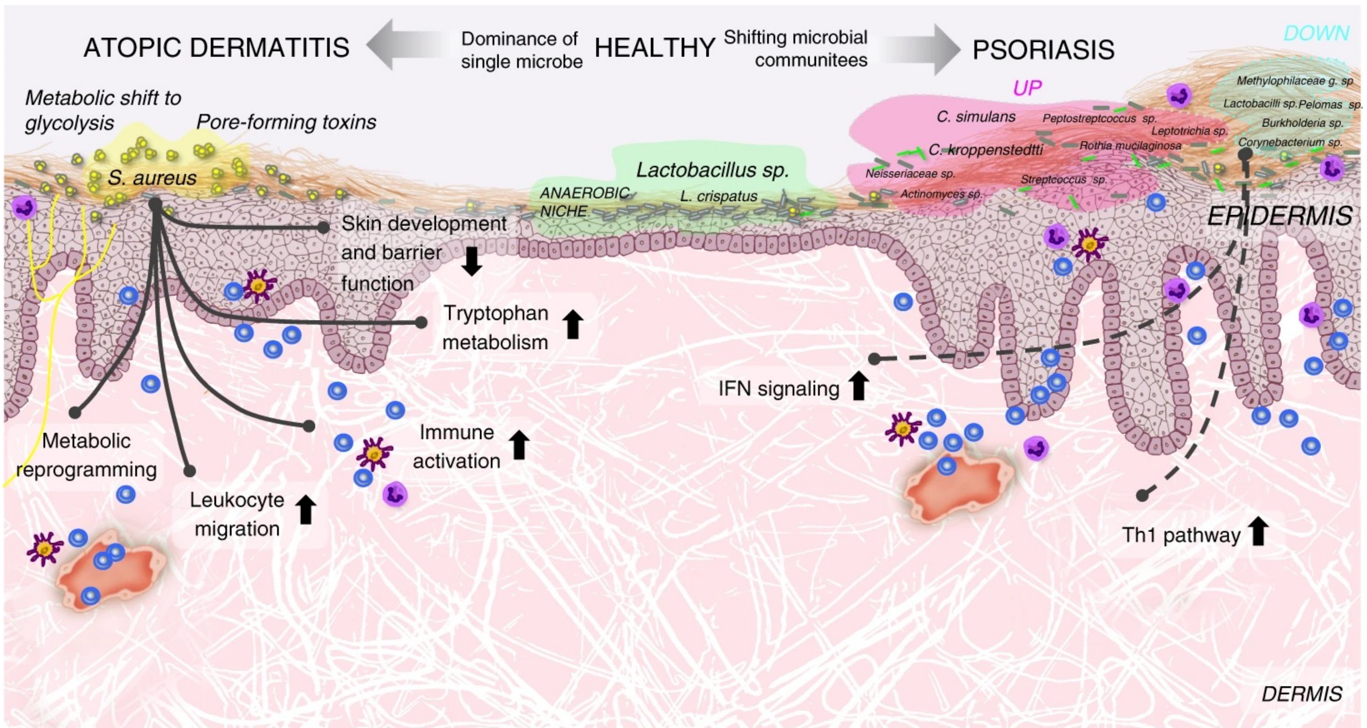
Gut microorganisms produce a repertoire of metabolites

Some of these, such as short chain fatty acids, can act as signaling molecules
→ promotes homeostasis & anti-inflammatory phenotype

Can influence susceptibility to immune-related disorders

Image: Rooks & Garrett, 2016

The skin microbiome plays a role in atopic dermatitis & psoriasis



Atopic dermatitis

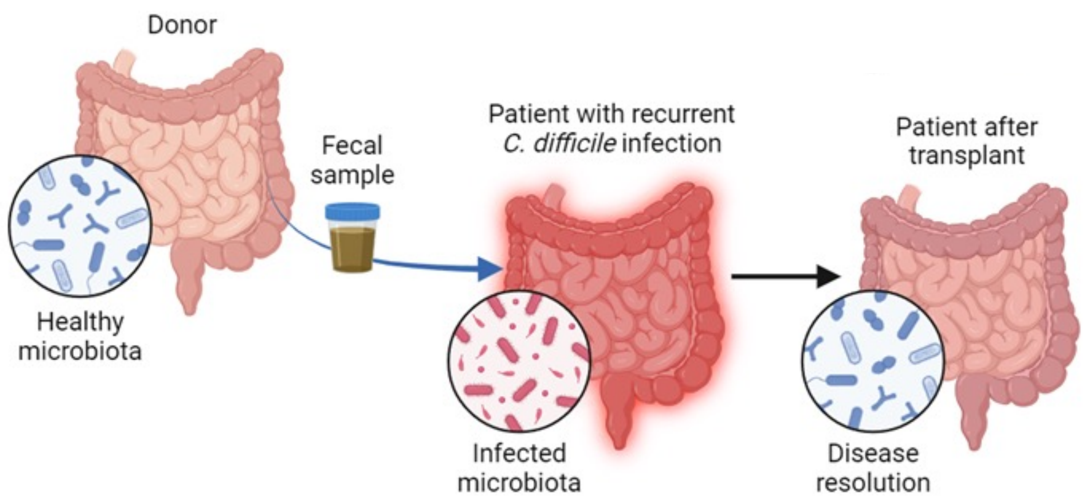
Overgrowth of *S. aureus* (usually commensal) ⟶
Loss of microbial diversity ⟶
Dysregulation of genes

Psoriasis

Colonization of multiple species

Image: Fyhrquist et al, 2019

Fecal microbiota transplants (FMT) are an emerging therapy



Used to treat:
Recurrent C. difficile infections

Ulcerative colitis, inflammatory bowel disease, Crohn's disease → associated with loss of diversity

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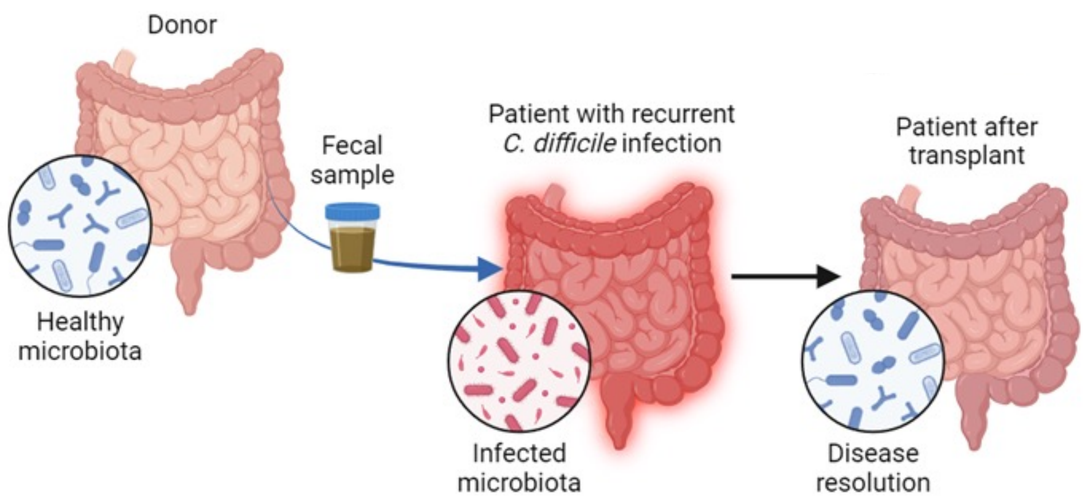

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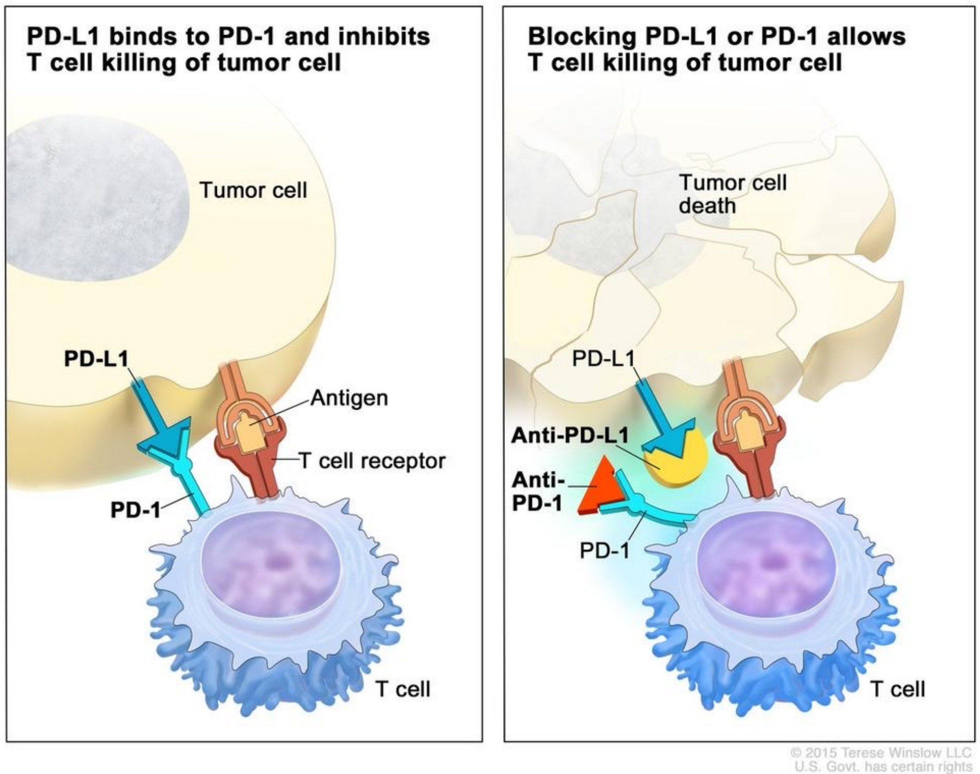
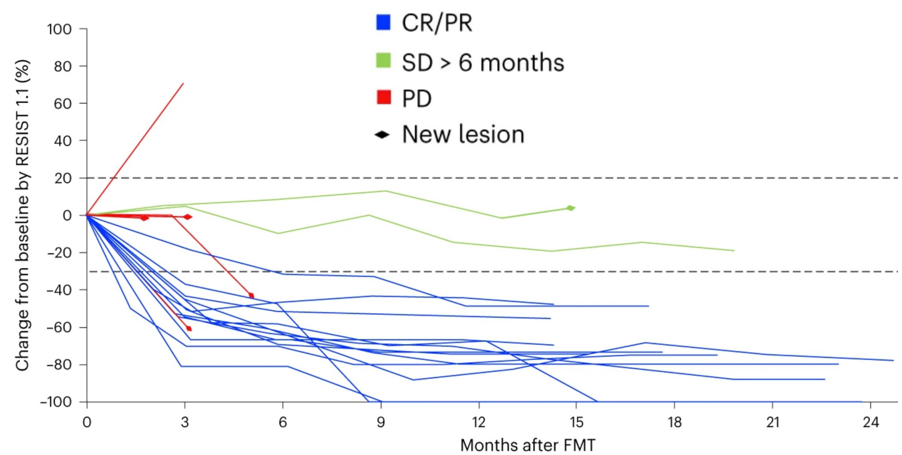
Not always sure how or why this works

FMT can overcome cancer immunotherapy resistance



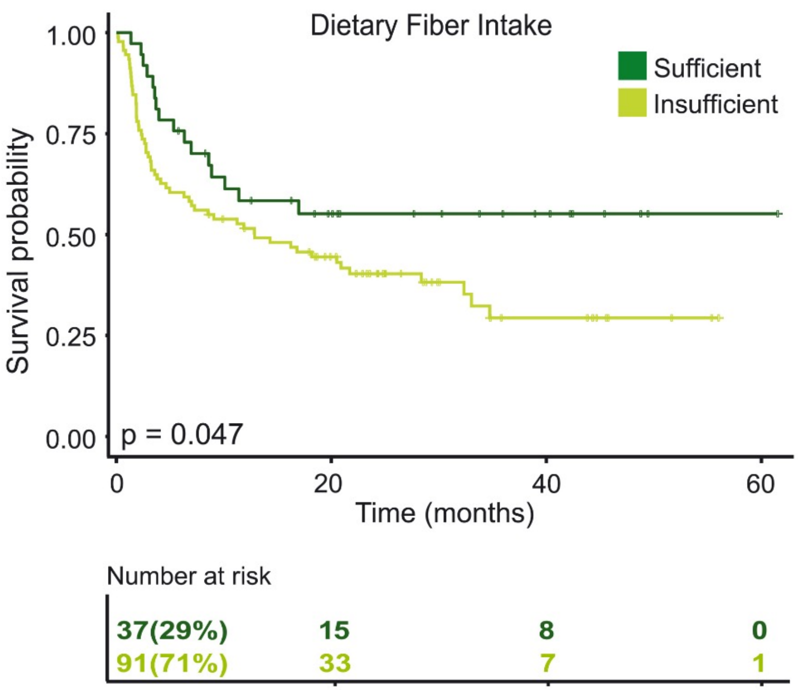


Gut microbiome plays a causal role in immune response

No specific taxa have been implicated

Images: NCI; Routey et al, 2023

Microbiome modulation can be used as a therapeutic intervention



Example: cancer

Patients on high fiber (prebiotic) diets have better response to immunotherapy

Dietary changes can help to reduce risk of developing colorectal cancer

Dietary fiber and probiotics influence the gut microbiome and melanoma immunotherapy response

CHRISTINE N. SPENCER, JENNIFER L. MCQUADE, VANCHESWARAN GOPALAKRISHNAN, JOHN A. MCCULLOCH, MARIE VETIZOU, ALEXANDRIA P. COGDILL, MD A. WADUD KHAN, XIAOTAO ZHANG, MICHAEL G. WHITE, [...], AND JENNIFER A. WARGO +79 authors Authors Info & Affiliations

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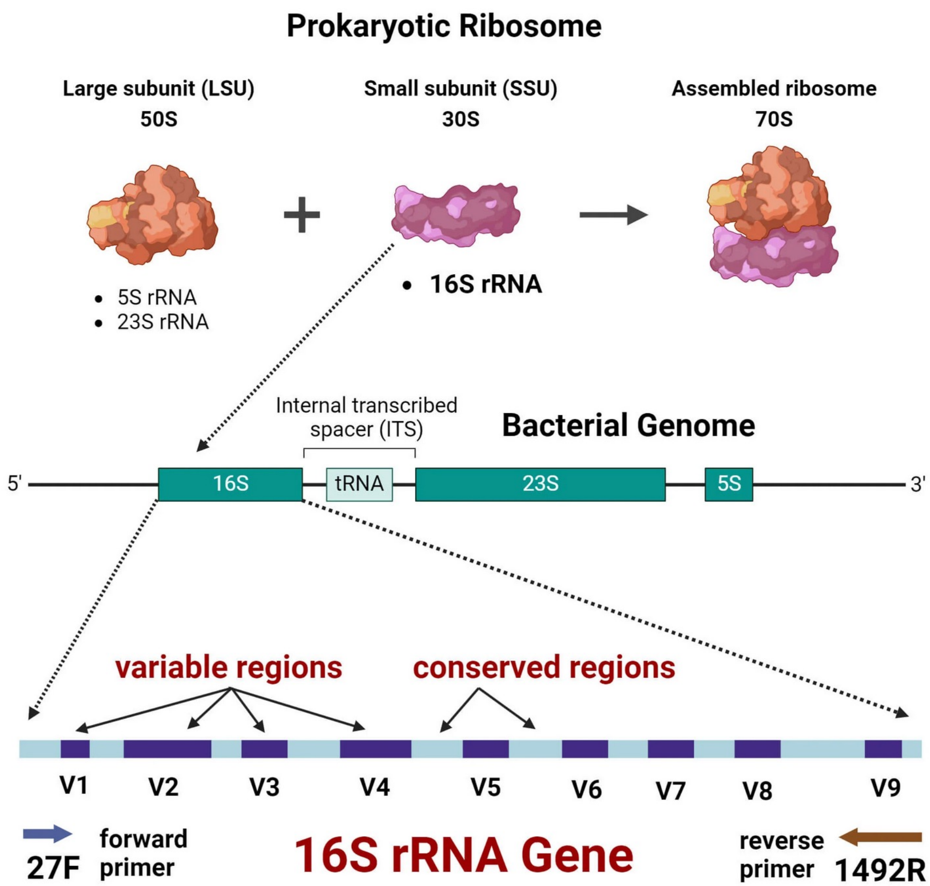
Sequencing methods: 16S rRNA



16S is a conserved marker gene
- Conserved regions allow for universal amplification
- Variable regions allow for identification

Can characterize taxonomic diversity of environmental samples

Cheaper but less thorough than metagenomic sequencing

Not easy to capture rare organisms

Image: MicrobeNotes

Sequencing methods: Whole-genome shotgun sequencing

Untargeted sequencing of all microbes in a sample

Can perform genome assemblies & obtain gene-level information

Allows for de novo and reference-based assembly

Bacteria are "one gene, one protein"
- No introns or alternative splicing
- Often predict proteins from 6-frame translated DNA

Image: Yang et al, 2021

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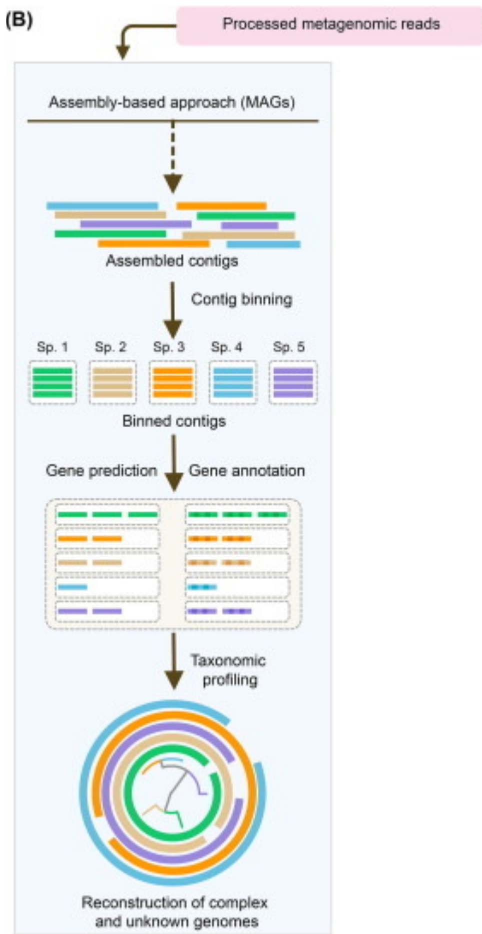

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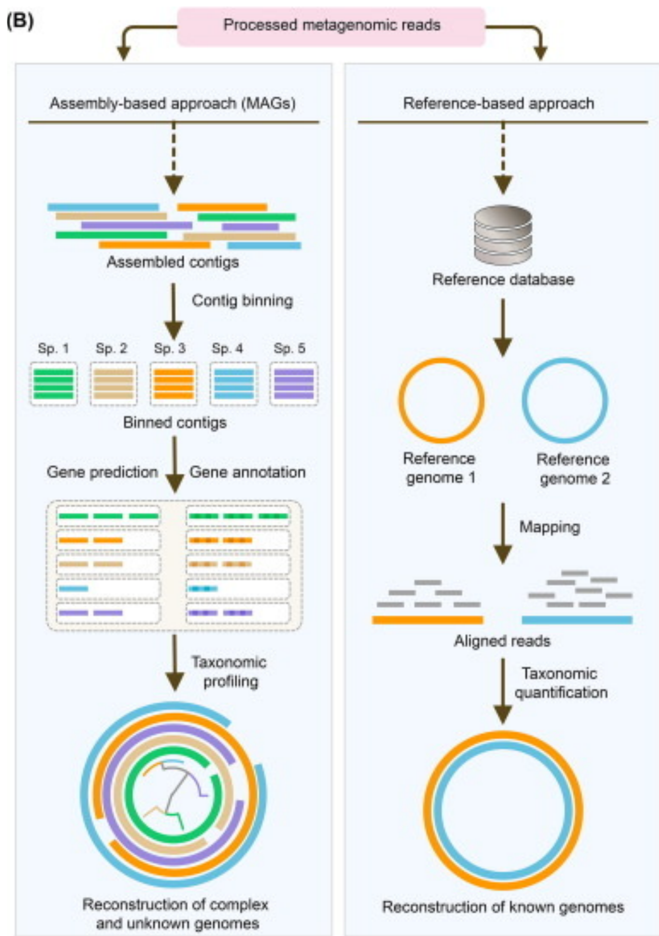

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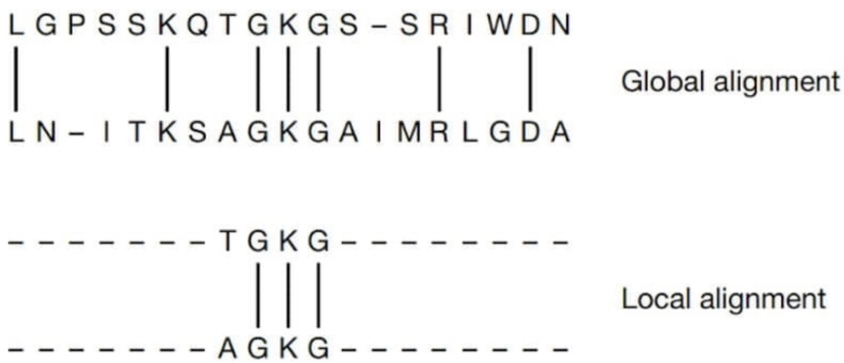
How do we quantify reads into functional categories?

Alignment



Alignments are very sensitive, but can be very slow

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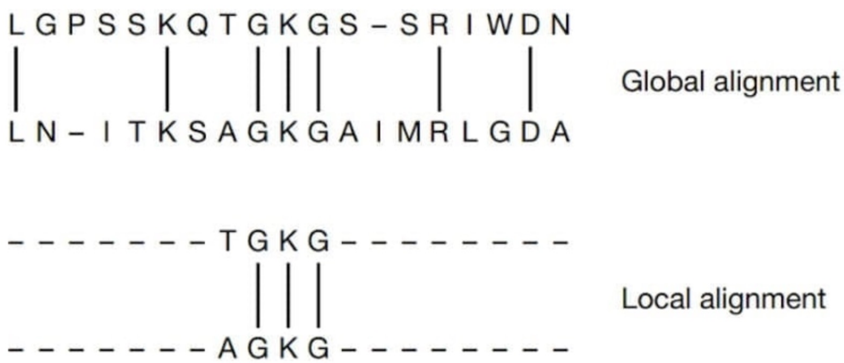

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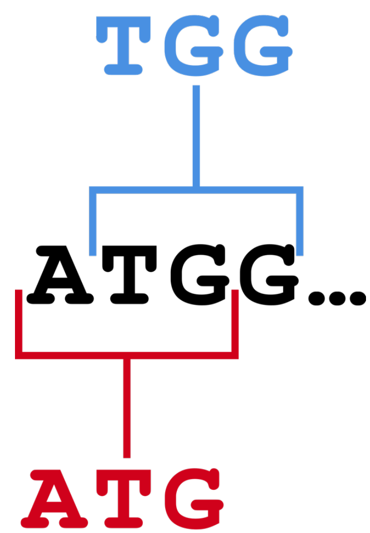
k-mer (k = 3)

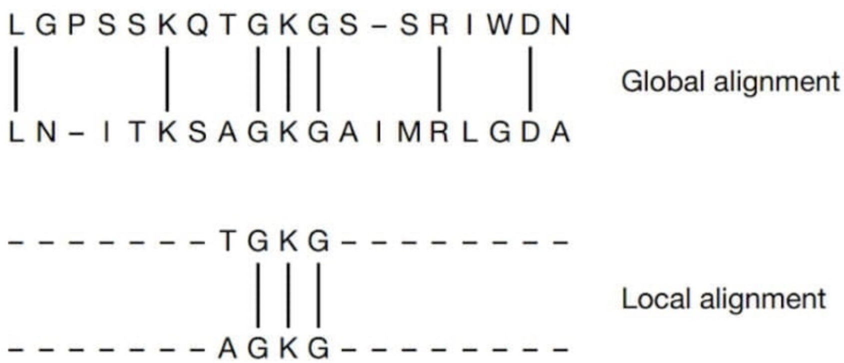


k-mer approaches are powerful and more resource efficient than alignments for sequence mapping

How do we quantify reads into functional categories?

Both approaches require comparison to some reference database of genes or proteins

Alignment



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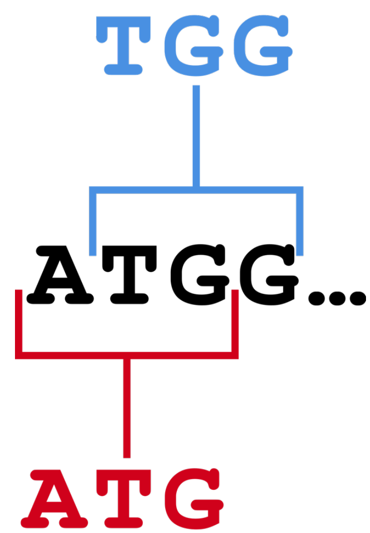

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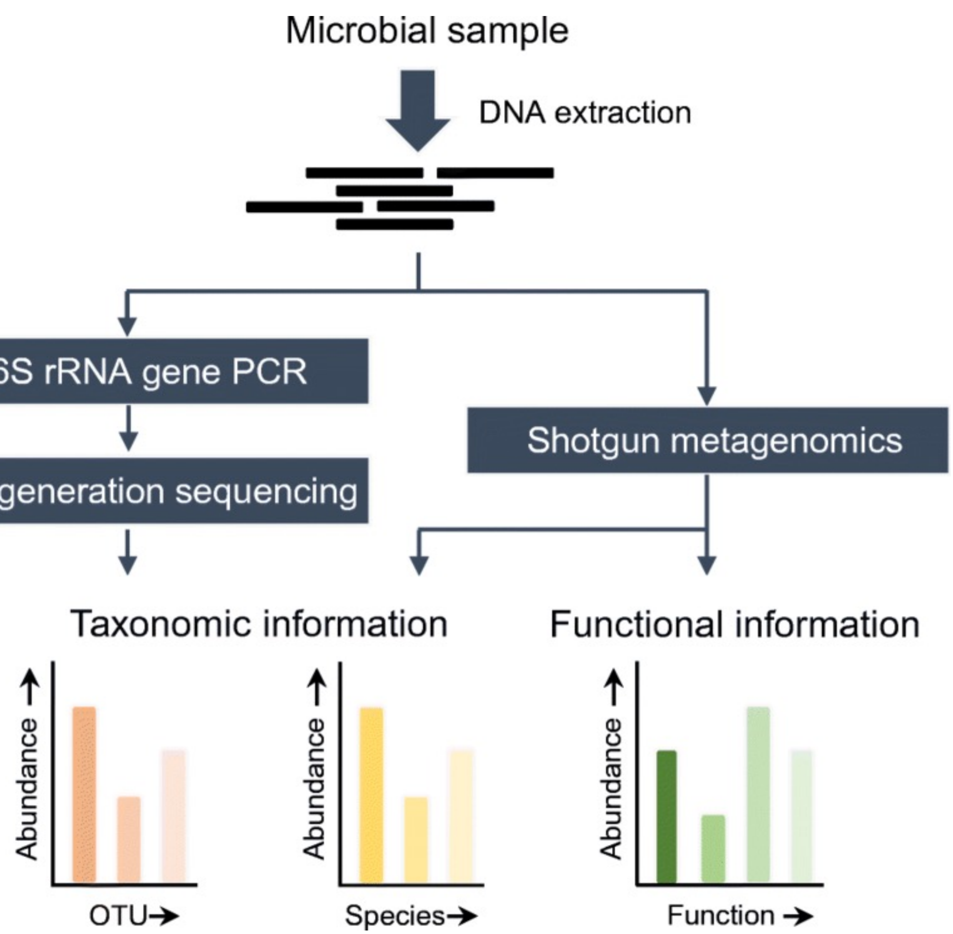
Sequencing methods: Why choose one over the other?



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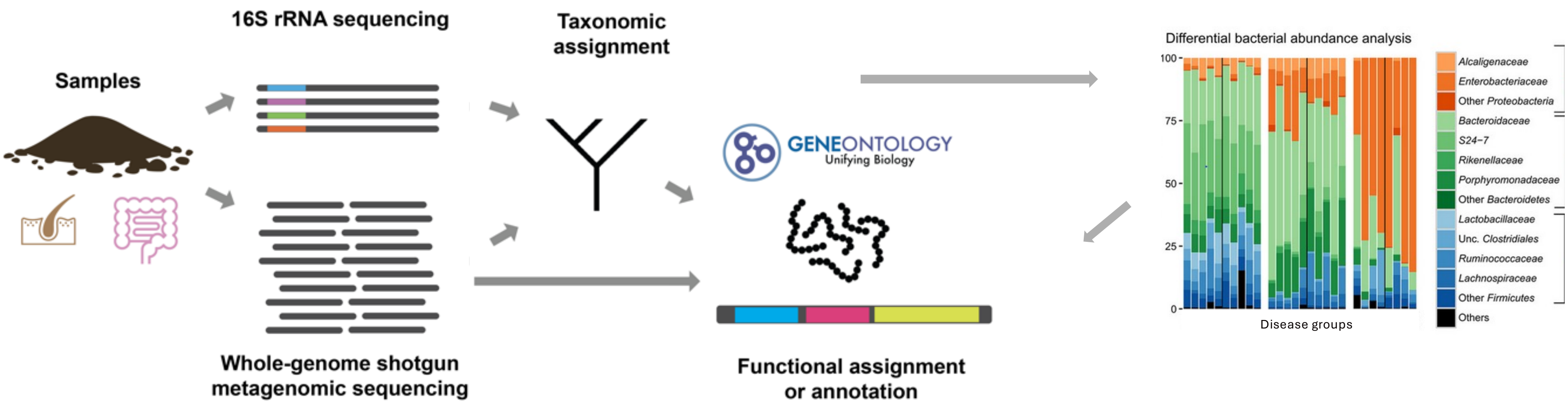
Typical metagenomic analysis workflow



Metagenomic sequencing quality control steps

1. Remove artifacts & low quality sequences
   Remove reads that align to the human genome, especially for low-density samples, e.g. tumor samples

2. Assign sequences to units, e.g. operational taxonomic units (OTUs), based on sequence similarity
   Remove chimeras

3. Assign taxonomy, e.g. phylum/genus/species via alignment

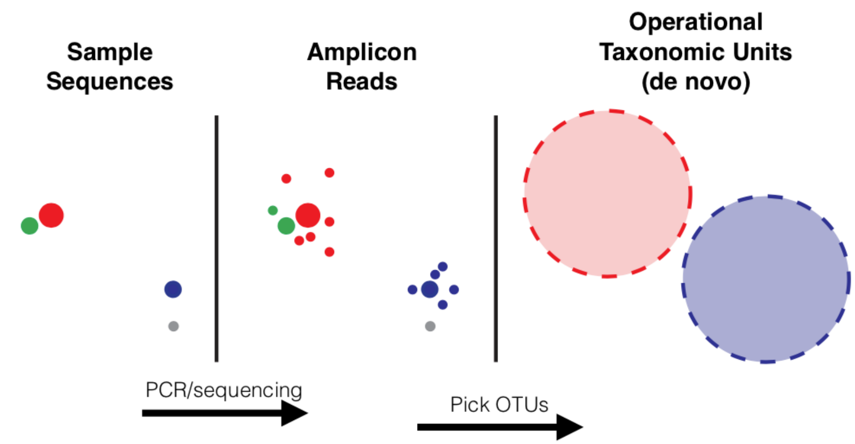
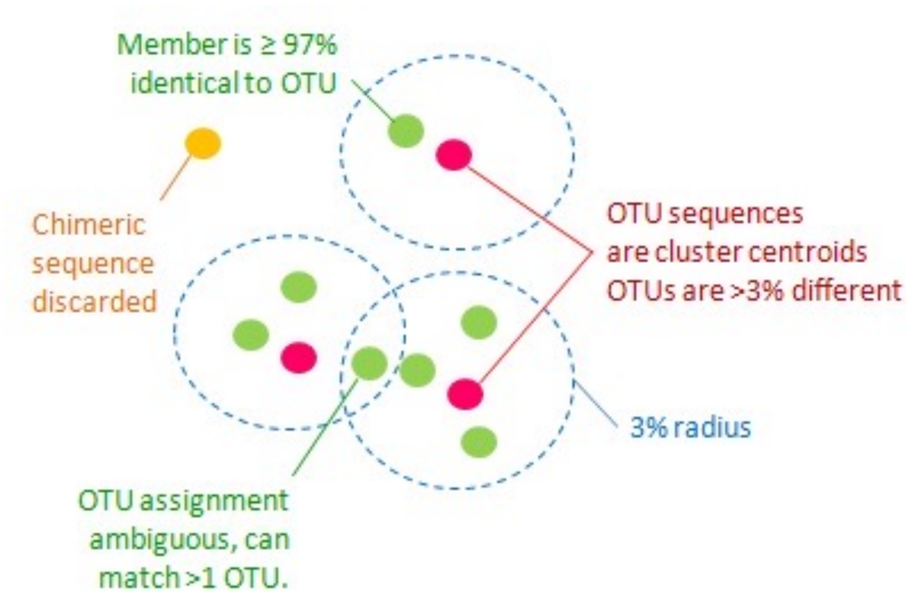


Image: Callahan et al, 2016

Operational Taxonomic Units (OTUs) help to group sequences



Used to classify groups of related individuals by sequence similarity (usually 97-100%)

Usually proxies for species

Most common unit of diversity for analyzing 16S marker gene sequence datasets

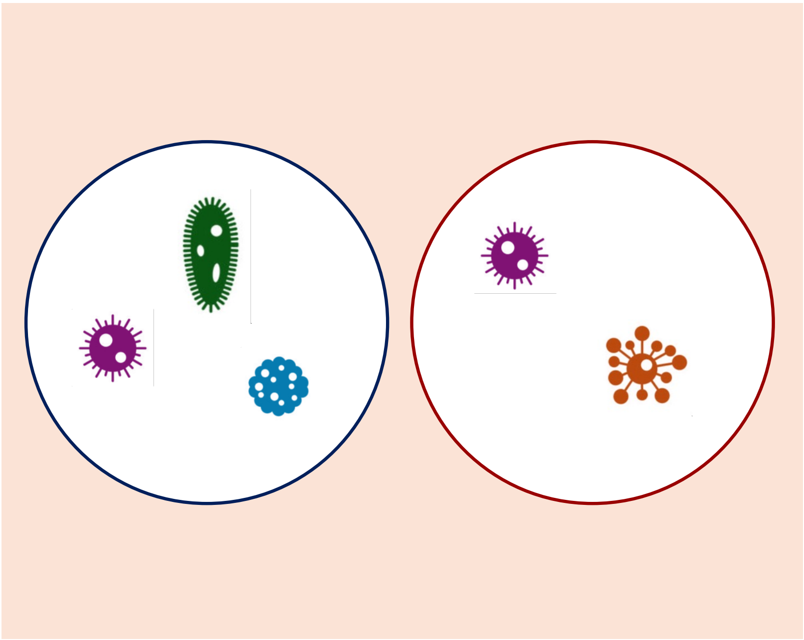
Can be inflated due to sequencing errors

Image: Edgar, 2013

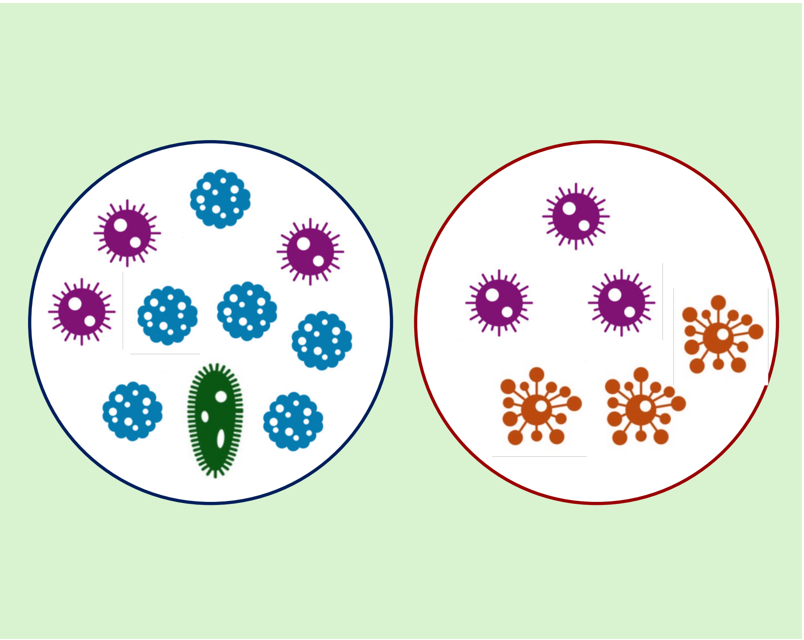
Microbiome data analysis: Alpha diversity

Diversity of species in a single sample

Count of species (Richness)



Evenness of species



Microbiome data analysis: Shannon diversity index (alpha)

$$\mathrm{H}(X) := -\sum_{x \in \mathcal{X}} p(x) \log p(x)$$

$p(x)$ = proportion of the *x*th OTU
$\mathcal{X}$ = total number of OTUs

Shannon entropy, Shannon-Weiner Index, Shannon-Weaver Index

Takes the proportional abundances of species into account

Measures expected amount of information needed to describe the state of a variable when considering the distribution of probabilities across all potential states

Originally taken from the field of information theory—concept of entropy is common in machine learning

Other alpha diversity metrics: Simpson index

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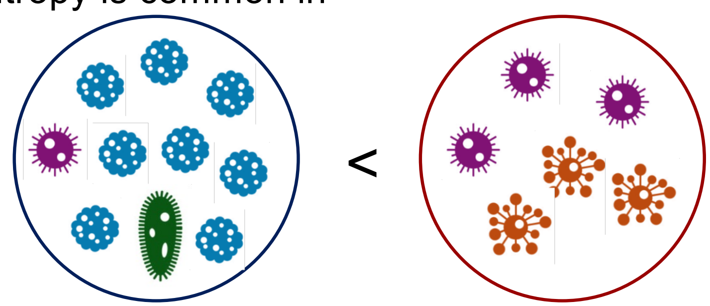
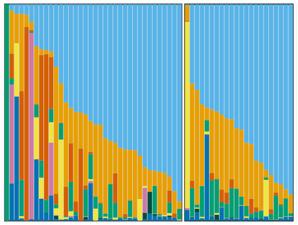
= -1* (.5*log(.5) + .5*log(.5))
= 0.69

= -1* (.1*log(.1) + .1*log(.1)) + .8*log(.8))
= 0.64



Microbiome data analysis: Beta diversity



Distance between samples

Presence-absence based

Jaccard distance
between samples *A* and *B*

$$1 - \frac{|A \cap B|}{|A \cup B|} = 1 - \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

Sorenson distance

$$\frac{B + C}{2A + B + C}$$

average fraction of unique species in a sample
B = unique from sample 1
C = unique from sample 2
A = intersecting species

Abundance based

Bray-Curtis dissimilarity

$$1 - \frac{2C_{ij}}{S_i + S_j}$$

S_i and S_j = total number of species in samples *i*, *j*

C_{ij} = sum lesser values for intersecting species in *i*, *j*

Manhattan distance

$$\sum_{i=1}^{n} | p_i - q_i |$$

p_i and q_i = abundance of species *i* in samples *p* and *q*

Distance metrics are not specific to microbiome data analysis

Microbiome data analysis: UniFrac distance (beta, phylogeny-based)

Considers phylogenetic relationships

Branches leading to taxa in both samples are considered shared

$$\left(\frac{sum\ of\ unshared\ branch\ lengths}{sum\ of\ all\ tree\ branch\ lengths}\right) =$$

$$fraction\ of\ total\ unshared\ branch\ lengths$$

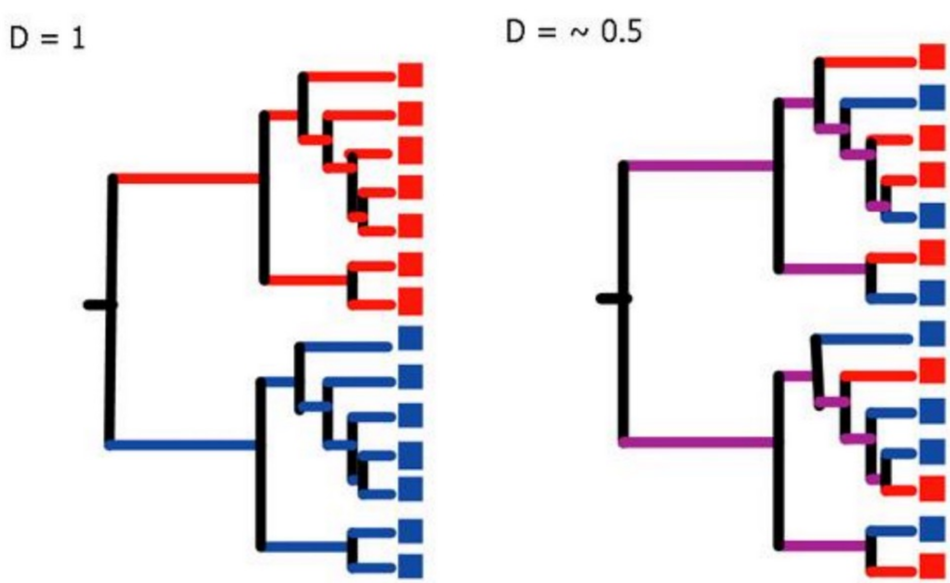


Image: Kyle Bittinger Microbiome Data Science Course

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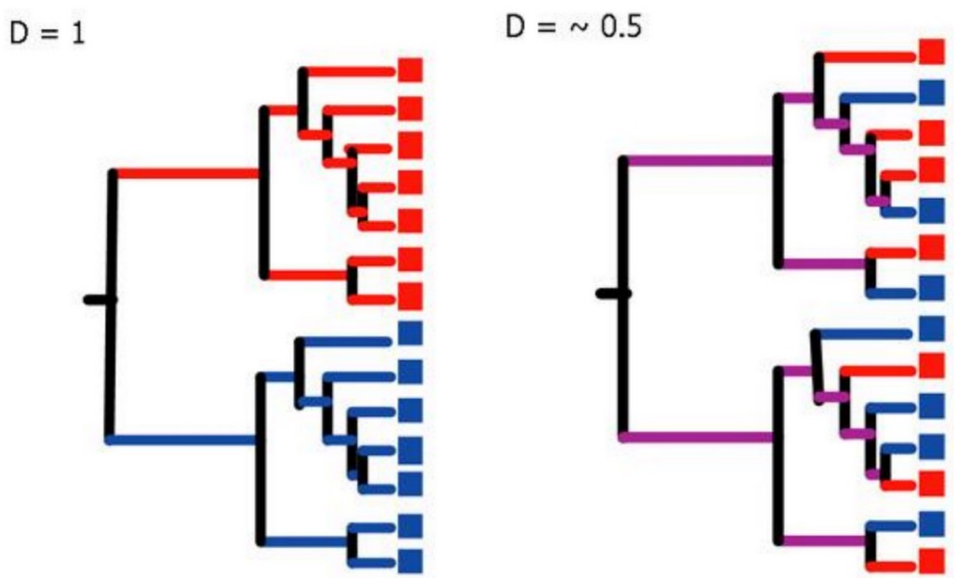


Unweighted

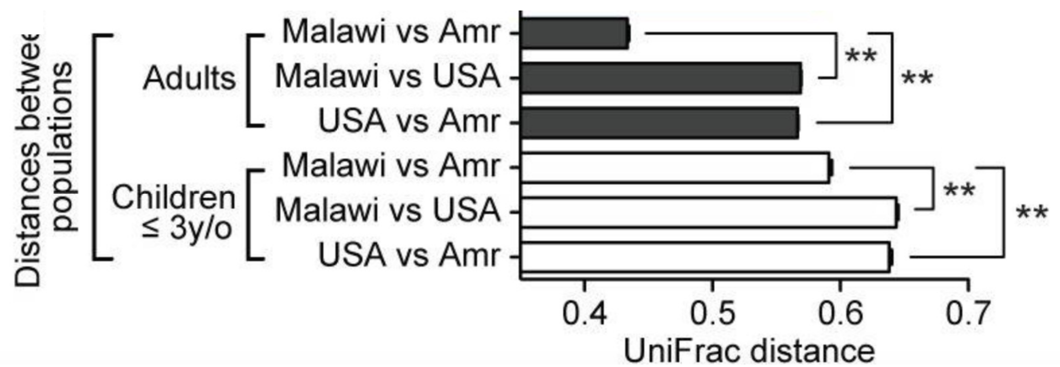
~ Jaccard + phylogeny emphasizes rare units

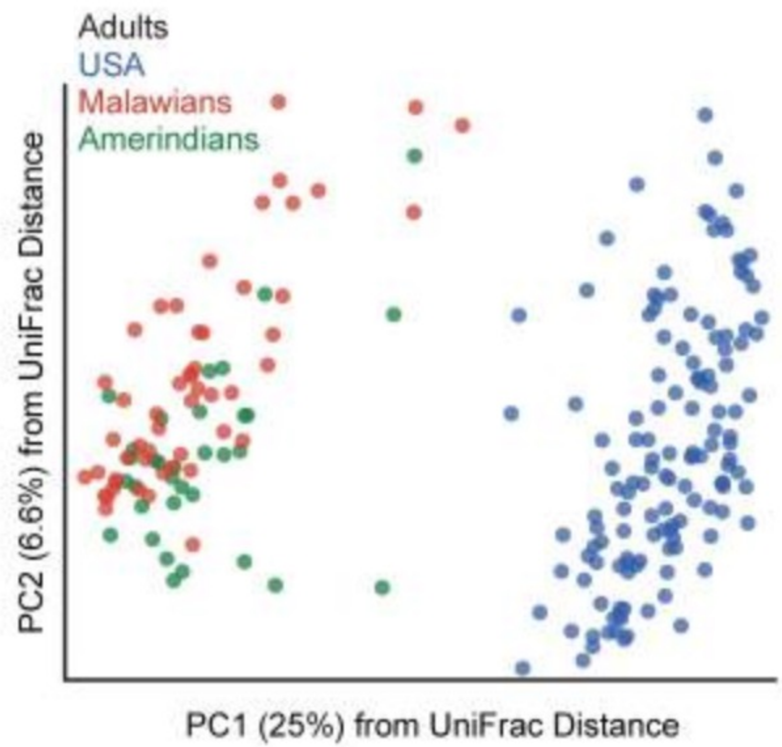
Weighted

~ Bray-Curtis + phylogeny emphasizes dominant units

Image: Kyle Bittinger Microbiome Data Science Course

Microbiome data analysis: Comparing diversity between populations





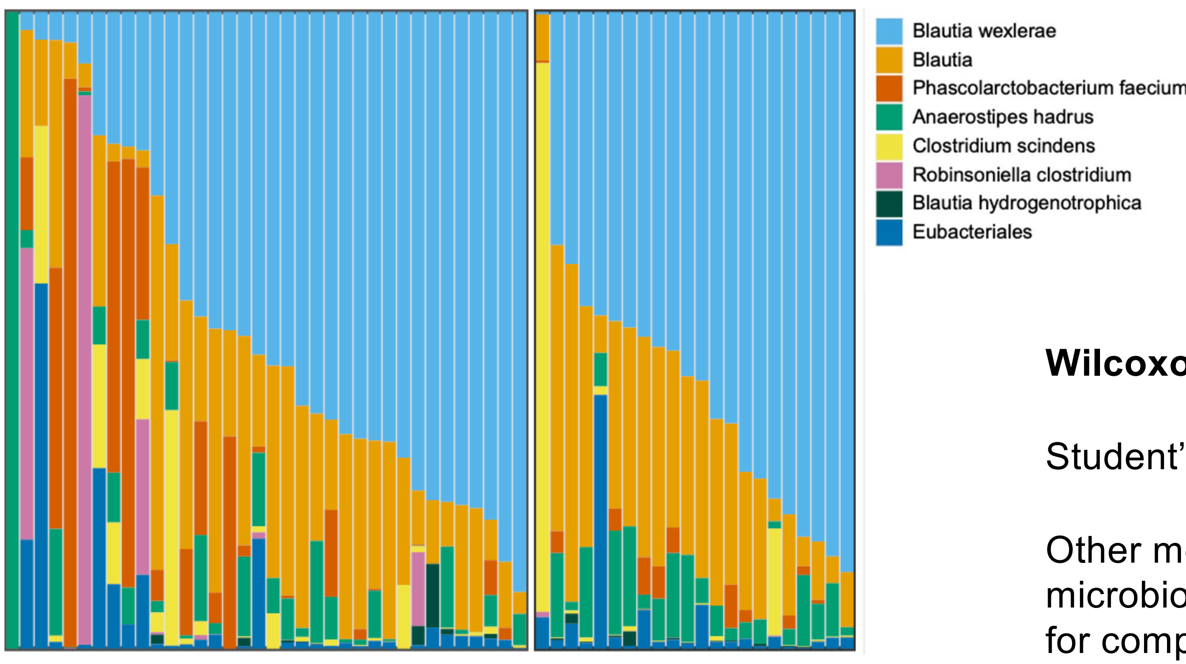
Can compare distances between disease groups, treatment response groups, populations, etc.

Can use any distance metrics

Images: Yatsunenko et al, 2012

Microbiome data analysis: Differential relative abundance

Which species or OTUs are differential between groups?



Disease   No Disease

Research | Open access | Published: 19 August 2022

A comprehensive evaluation of microbial differential abundance analysis methods: current status and potential solutions

Lu Yang & Jun Chen

Microbiome **10**, Article number: 130 (2022) | Cite this article

23k Accesses | **60** Citations | **28** Altmetric | Metrics

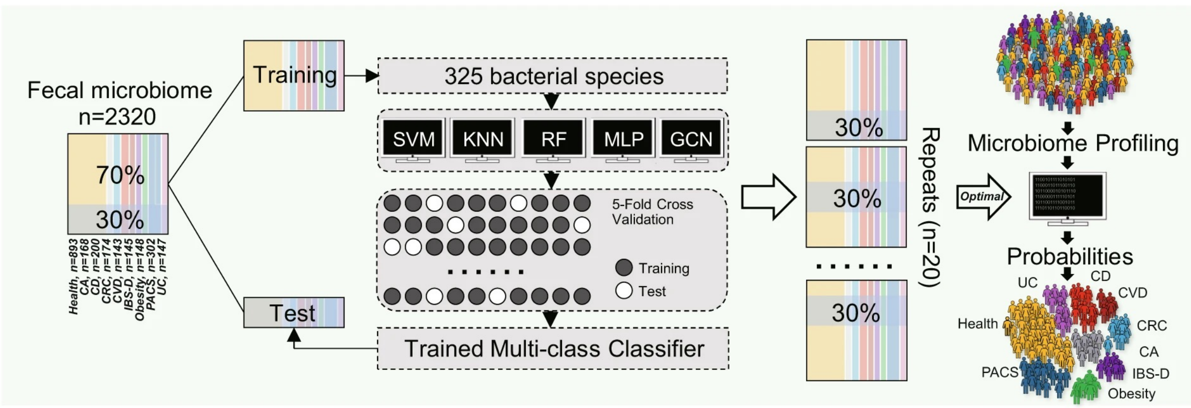
Wilcoxon rank-rum test

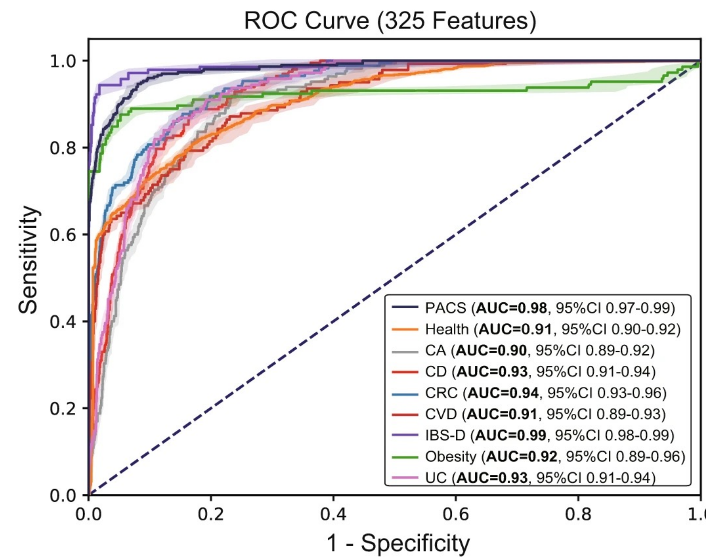
Student’s t-test

Other methods that are specific to microbiome, e.g. ANCOM which accounts for compositionality

Microbiome data analysis: Classification

Which species or OTUs can differentiate between groups?





Can use any machine learning method

Features are normalized OTU/species/gene counts

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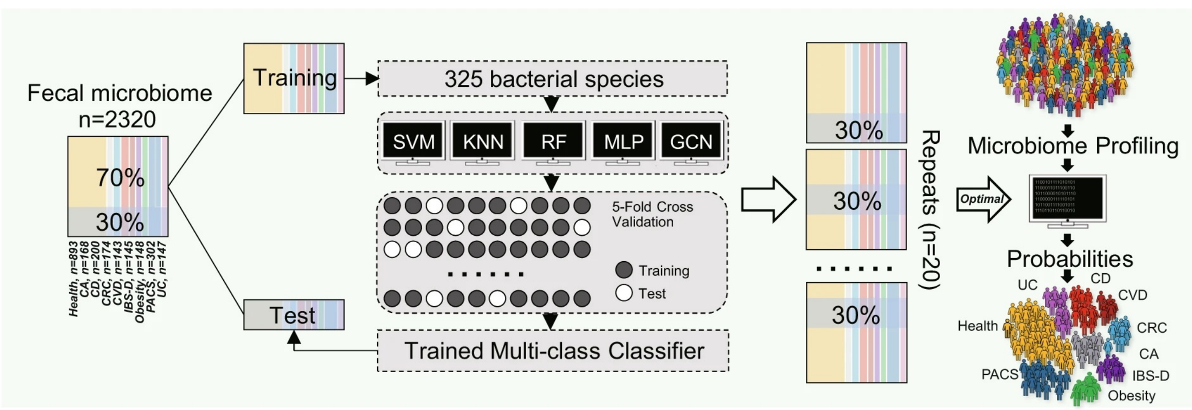
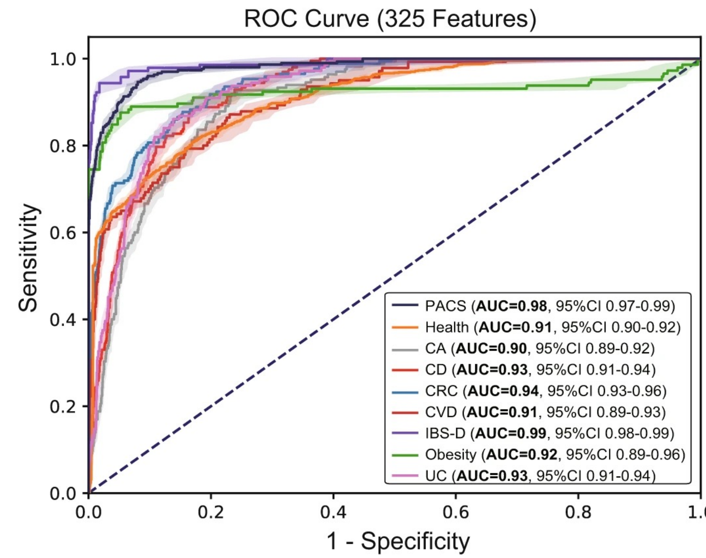


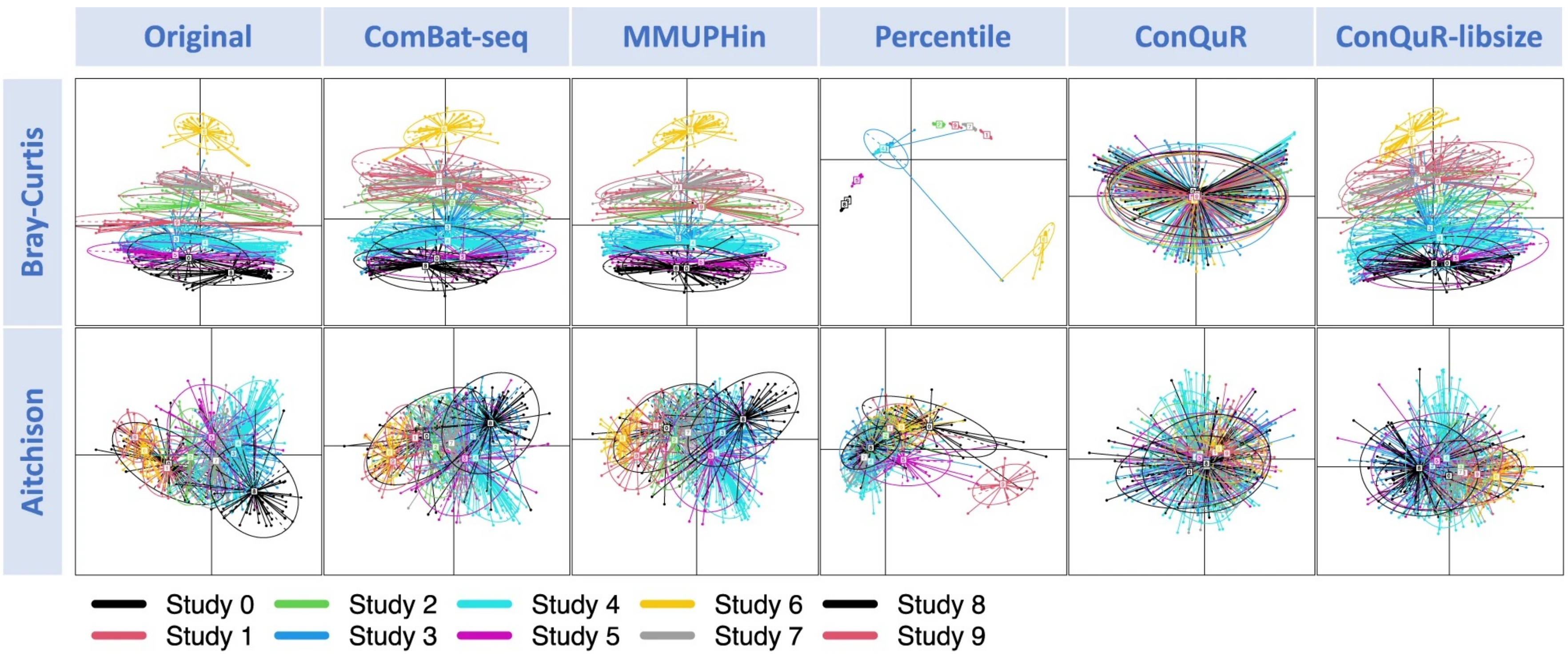


Can use any machine learning method

Features are normalized OTU/species/gene counts

What to do with these results?

Analysis considerations: Experimental design & batch effects



Images: Ling et al, 2022

Analysis considerations: Quality control

Microbiome studies are prone to human read contamination

bioRxiv [Preprint]. 2023 Jul 31:2023.07.28.550993. [Version 1] doi: 10.1101/2023.07.28.550993

Major data analysis errors invalidate cancer microbiome findings

Abraham Gihawi [1,*], Yuchen Ge [2,3,*], Jennifer Lu [2,3,*], Daniela Puiu [2,3], Amanda Xu [2], Colin S Cooper [1], Daniel S Brewer [1,4], Mihaela Pertea [2,3,5], Steven L Salzberg [2,3,5,6,†]

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PMCID: PMC10418105 PMID: 37577699

Article | Open access | Published: 23 February 2024

Robustness of cancer microbiome signals over a broad range of methodological variation

Gregory D. Sepich-Poore, Daniel McDonald, Evguenia Kopylova, Caitlin Guccione, Qiyun Zhu, George Austin, Carolina Carpenter, Serena Fraraccio, Stephen Wandro, Tomasz Kosciolek, Stefan Janssen, Jessica L. Metcalf, Se Jin Song, Jad Kanbar, Sandrine Miller-Montgomery, Robert Heaton, Rana Mckay, Sandip Pravin Patel, Austin D. Swafford, Tal Korem & Rob Knight

Oncogene **43**, 1127–1148 (2024) | Cite this article

16k Accesses | **6** Citations | **22** Altmetric | Metrics

Article | Published: 11 March 2020

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Nature **579**, 567–574 (2020) | Cite this article

107k Accesses | **676** Citations | **977** Altmetric | Metrics

This article was retracted on 26 June 2024

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Oncogene **43**, 1127–1148 (2024) | Cite this article

Article | Open access | Published: 15 May 2023

Reconstruction of the personal information from human genome reads in gut metagenome sequencing data

Yoshihiko Tomofuji, Kyuto Sonehara, Toshihiro Kishikawa, Yuichi Maeda, Kotaro Ogawa, Shuhei Kawabata, Takuro Nii, Tatsusada Okuno, Eri Oguro-Igashira, Makoto Kinoshita, Masatoshi Takagaki, Kenichi Yamamoto, Takashi Kurakawa, Mayu Yagita-Sakamaki, Akiko Hosokawa, Daisuke Motooka, Yuki Matsumoto, Hidetoshi Matsuoka, Maiko Yoshimura, Shiro Ohshima, Shota Nakamura, Hidenori Inohara, Haruhiko Kishima, Hideki Mochizuki, ... Yukinori Okada + Show authors

Nature Microbiology **8**, 1079–1094 (2023) | Cite this article

Article | Published: 11 March 2020

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Gregory D. Poore, Evguenia Kopylova, Qiyun Zhu, Carolina Carpenter, Serena Fraraccio, Stephen Wandro, Tomasz Kosciolek, Stefan Janssen, Jessica Metcalf, Se Jin Song, Jad Kanbar, Sandrine Miller-Montgomery, Robert Heaton, Rana Mckay, Sandip Pravin Patel, Austin D. Swafford & Rob Knight

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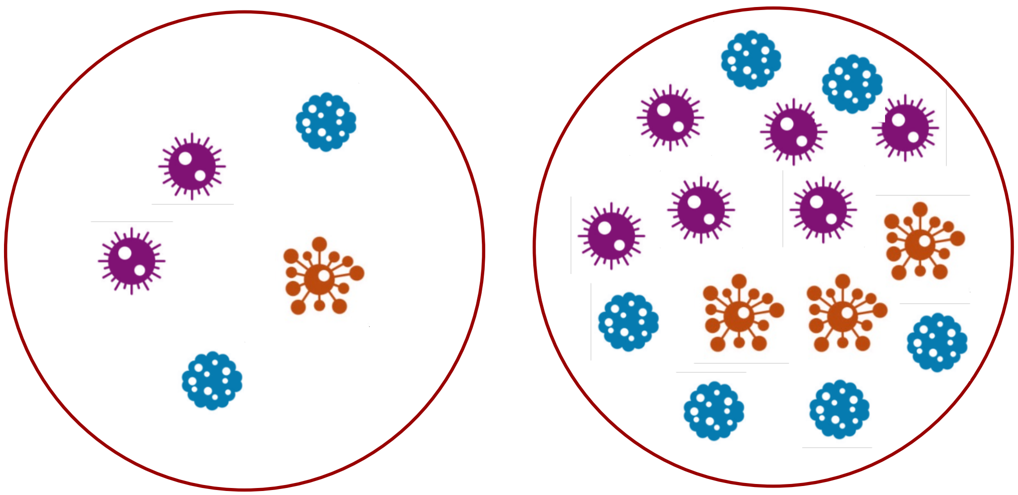
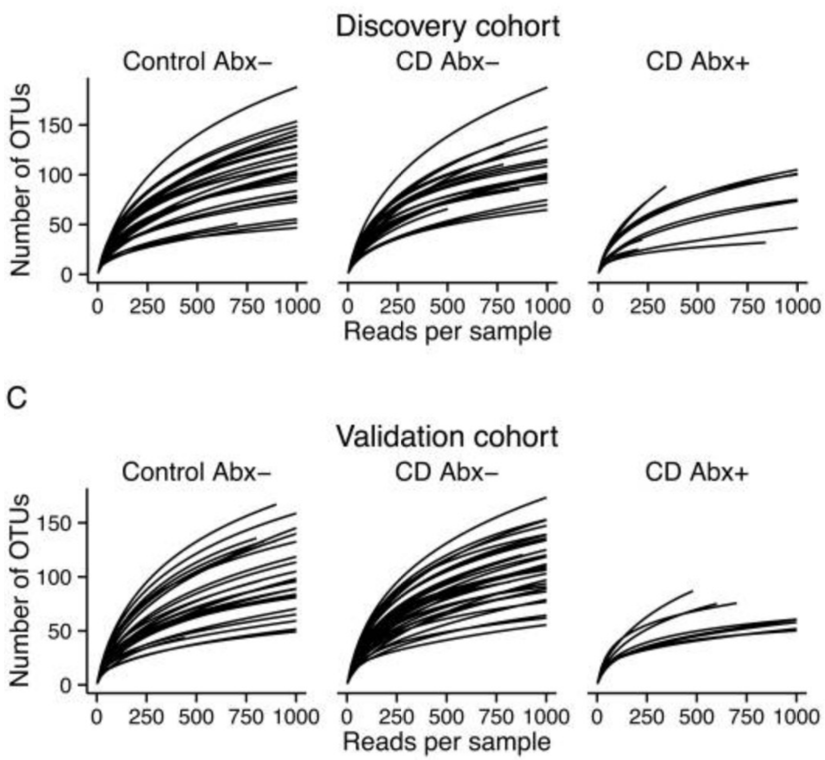
Analysis considerations: Compositionality

How do we tell the difference between these using alpha and beta diversity?



Analysis considerations: Sequencing depth



As the sequencing depth increases, you find more species

If you are comparing samples, need to make sure the depth is consistent

Does begin to plateau

Article | Open access | Published: 12 April 2018

Impact of sequencing depth on the characterization of the microbiome and resistome

Rahat Zaheer, Noelle Noyes, Rodrigo Ortega Polo, Shaun R. Cook, Eric Marinier, Gary Van Domselaar, Keith E. Belk, Paul S. Morley & Tim A. McAllister

Scientific Reports **8**, Article number: 5890 (2018) | Cite this article

26k Accesses | **14** Altmetric | Metrics

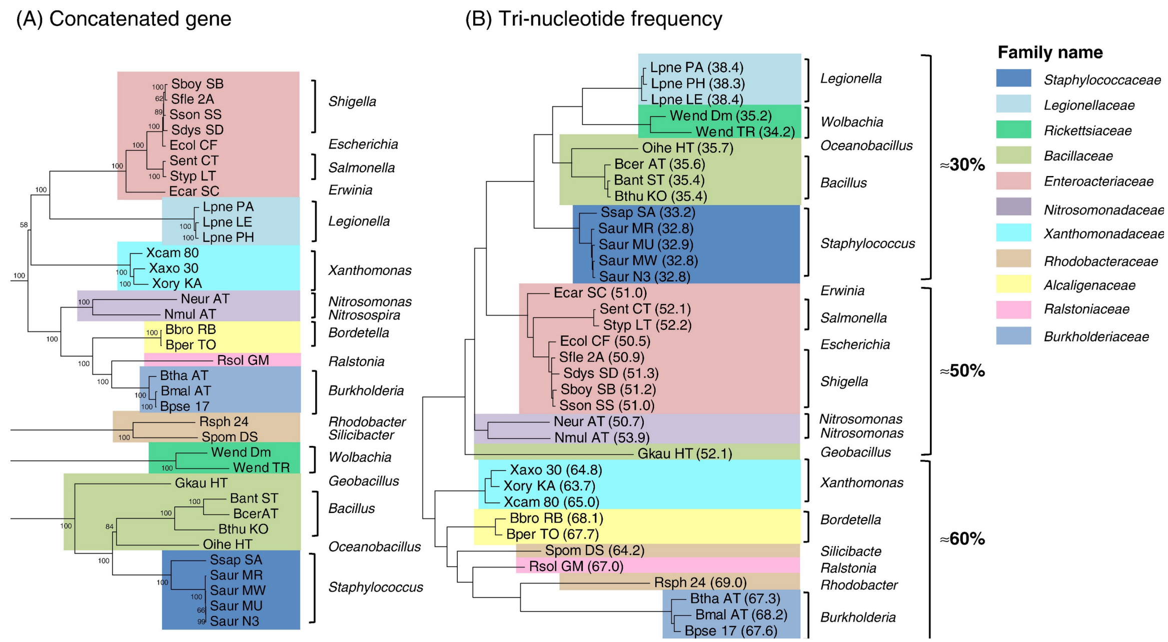
Images:Kelsen et al, 2016

Analysis considerations: Taxonomy

How do we choose a cutoff or level to use? genus, phylum, species

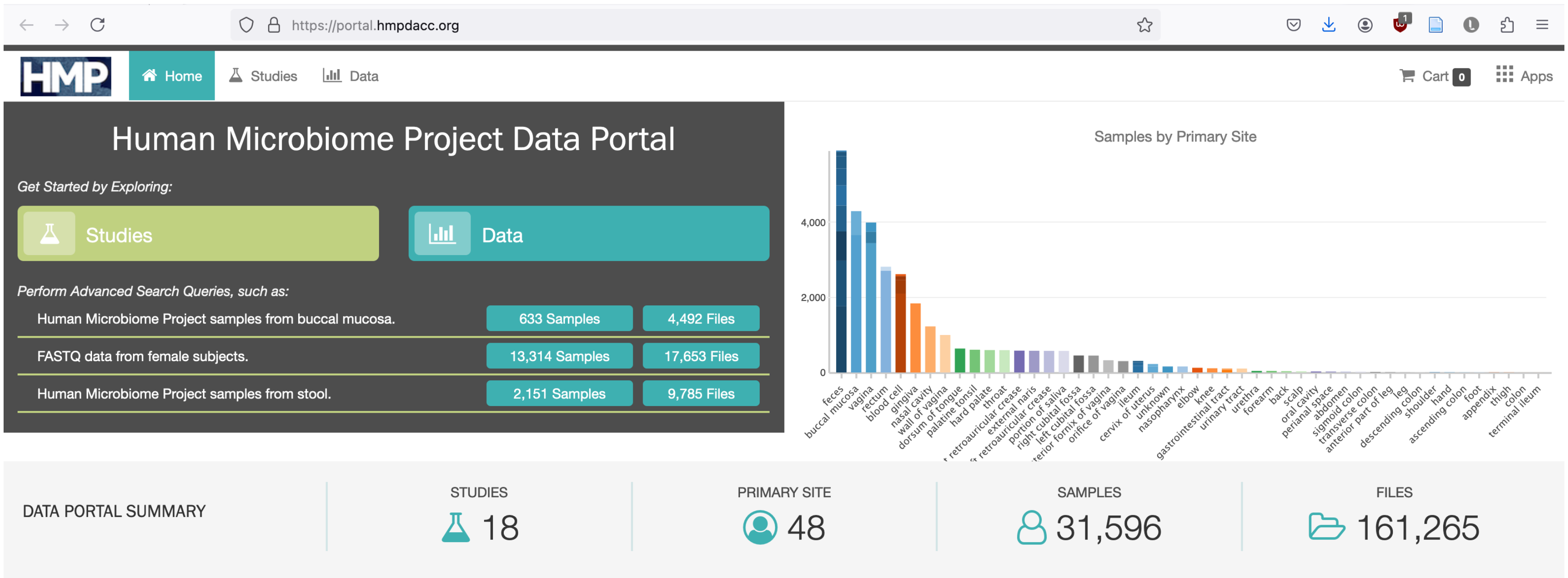
We expect taxa to be different in different populations

Does phylogeny relate to gene content?



Images: Takahashi, 2009

Important resources: The Human Microbiome Project

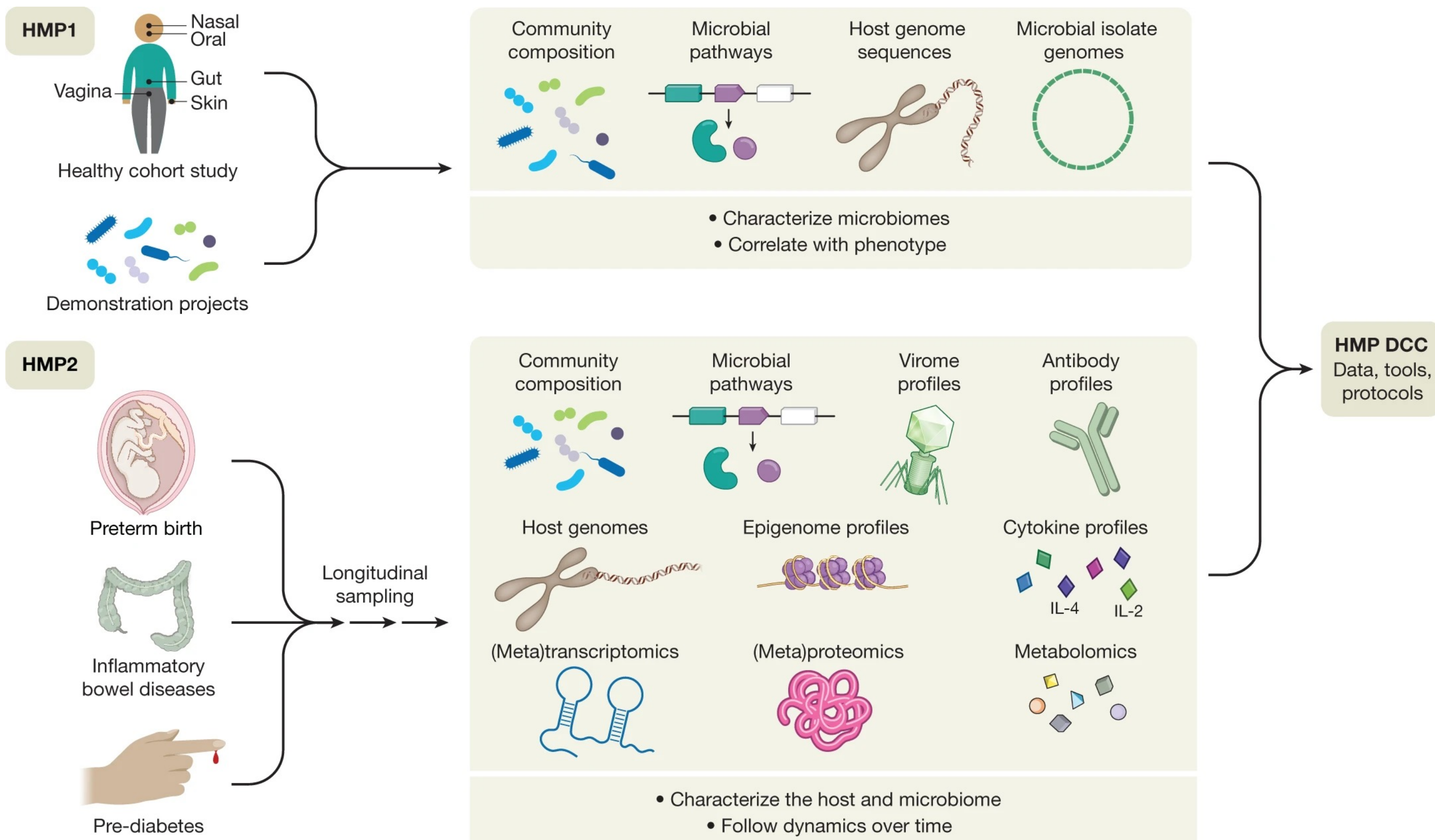


Multi-institutional Initiative from the NIH — search for a healthy “core” microbiome

16S & shotgun sequencing + whole-genome, metatranscriptomics, metabolomics, immunoproteomics

Multiple body sites

Important resources: The Human Microbiome Project



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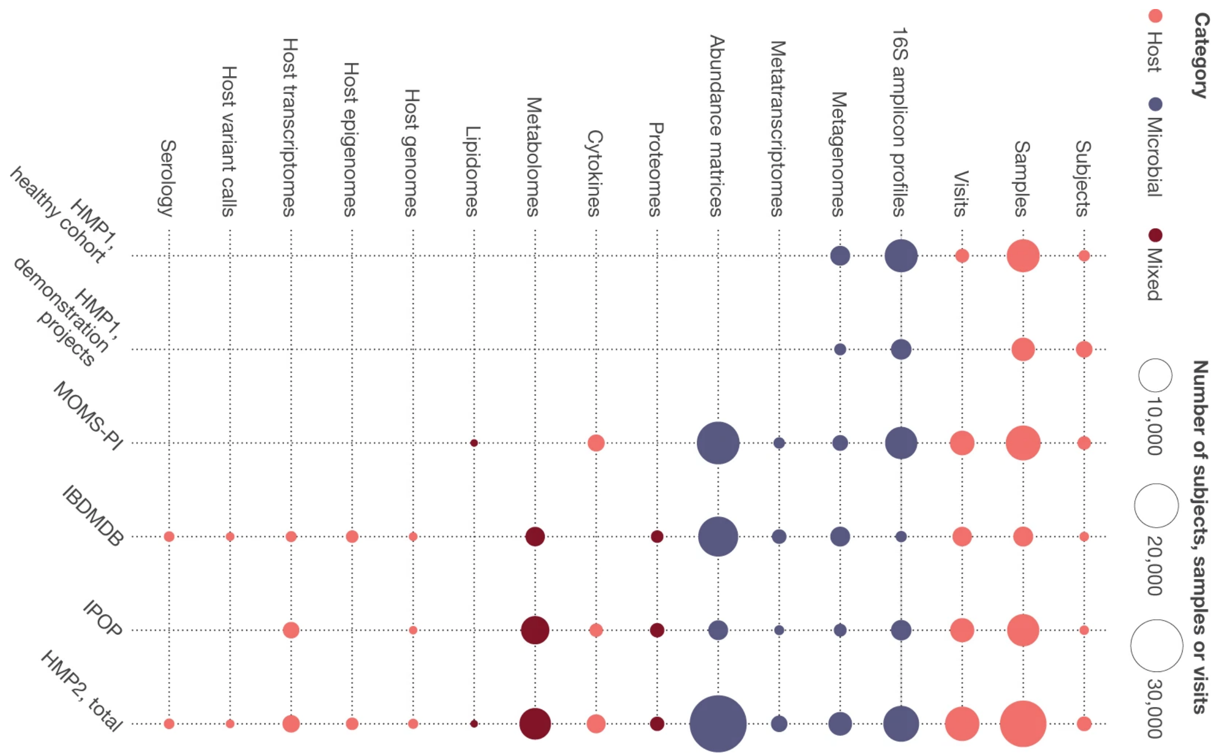

Important resources: Reference databases

How can reference databases affect microbiome analysis?

We can only align to known things

For de novo assembly, we still need to compare to something known

Future directions for microbiome research

Getting larger cohorts

Integrating with multi-omics

Overcome computational bottlenecks

Focus on genes, proteins, pathways rather than taxonomy

Mechanisms rather than associations

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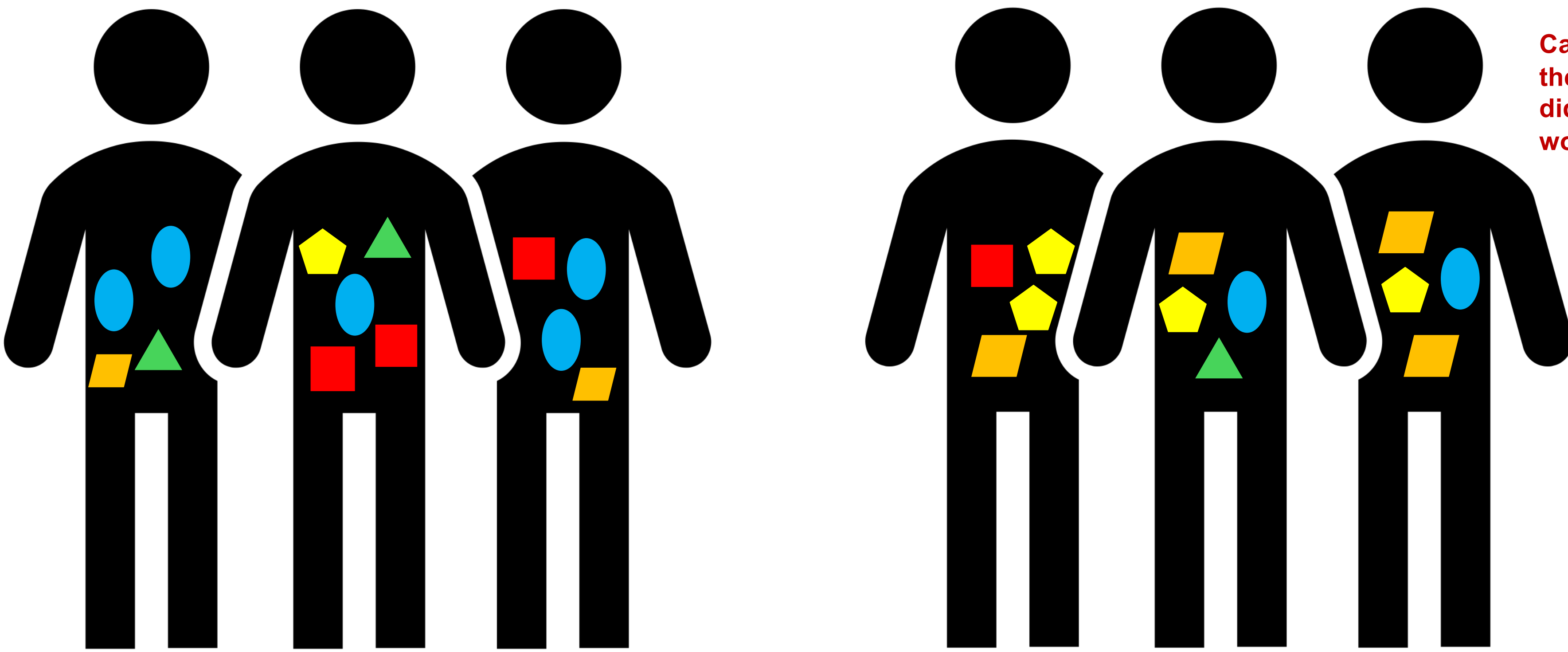
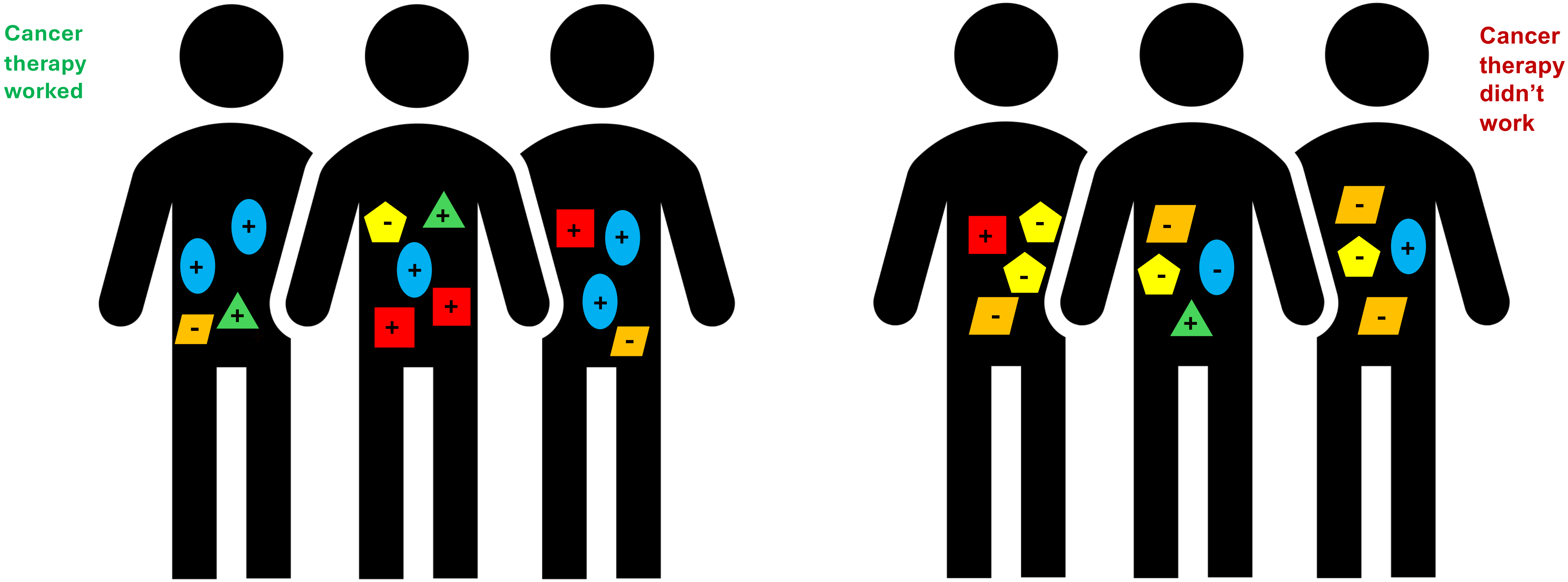
Which bacteria is causing people to respond to the drug?

Cancer therapy worked

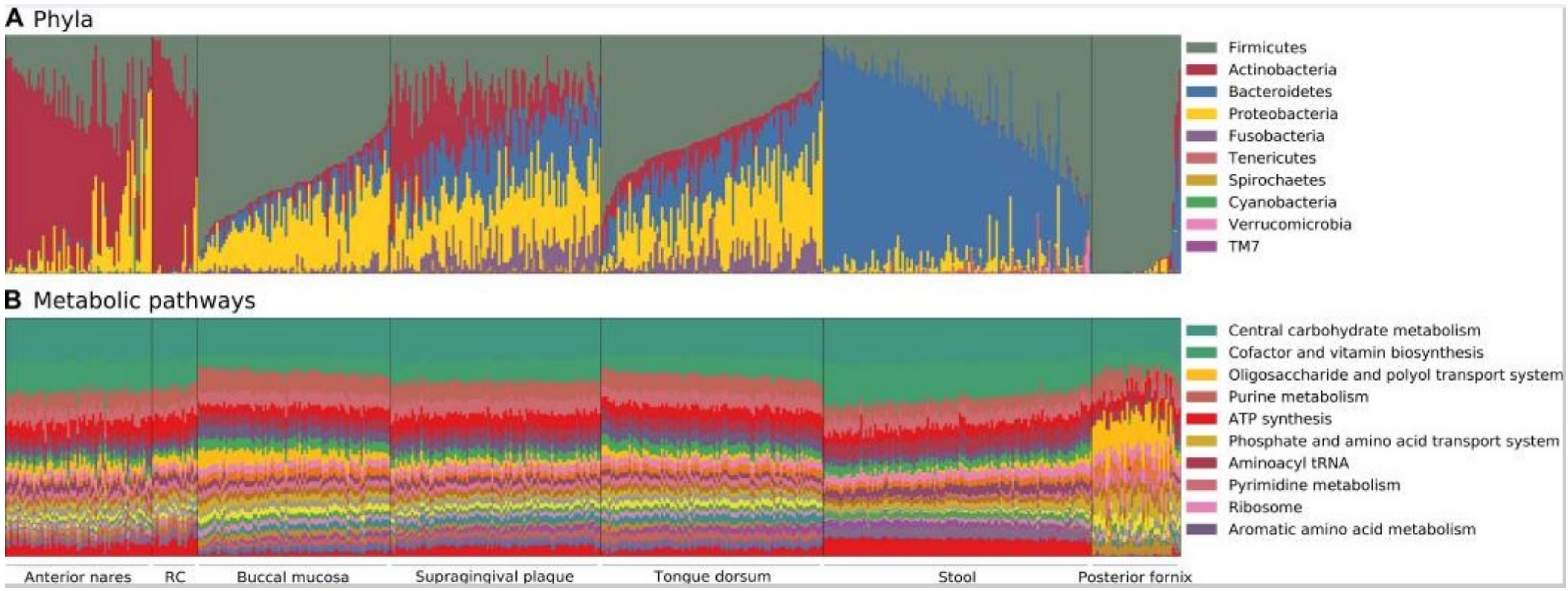


Cancer therapy didn't work

Which bacteria is causing people to respond to the drug?

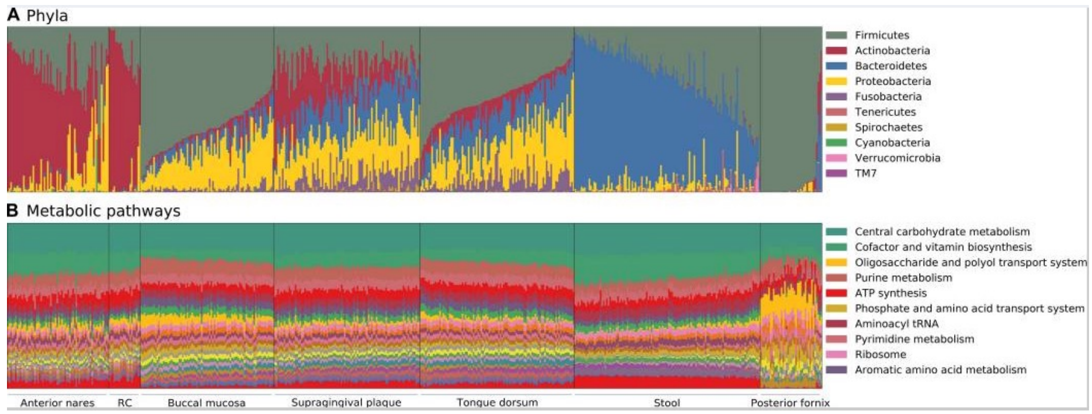


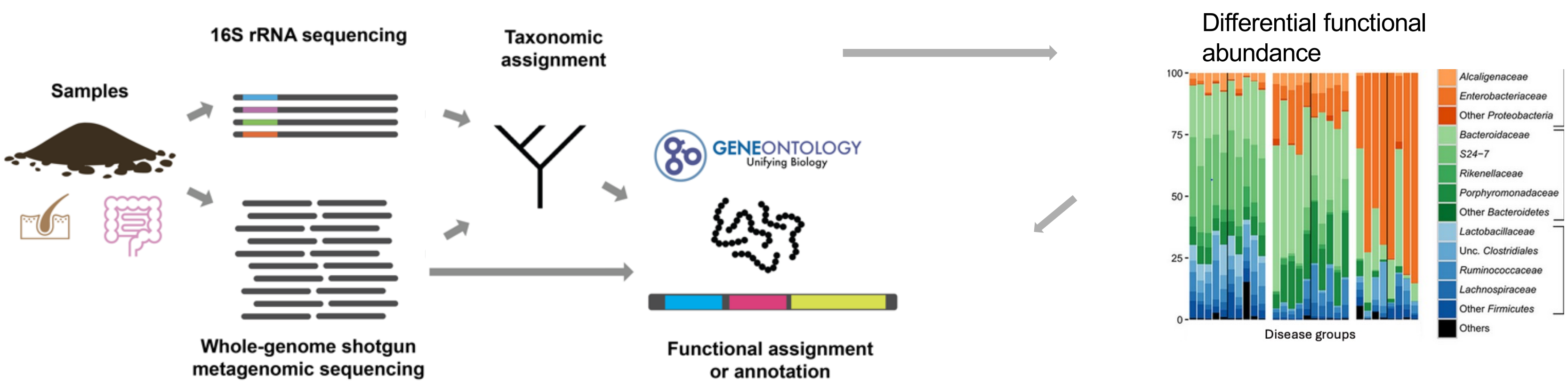
The gut microbiome is more stable functionally than taxonomically



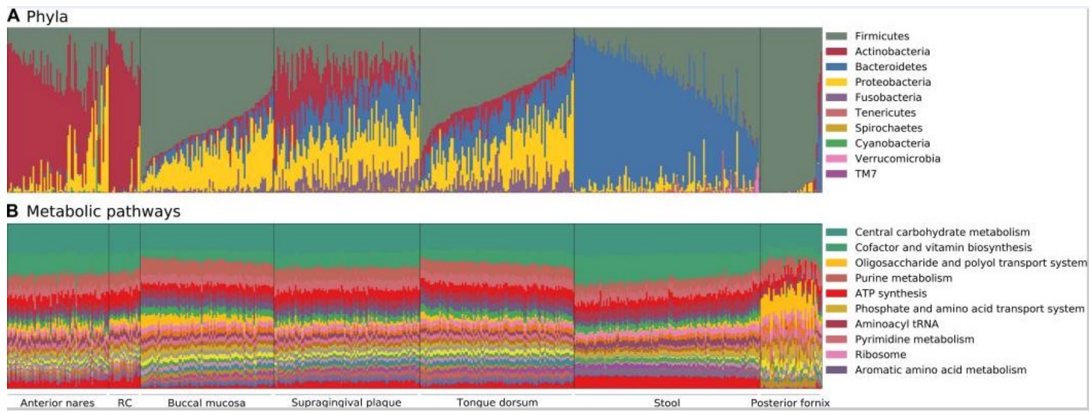
Images: Human Microbiome Project Consortium

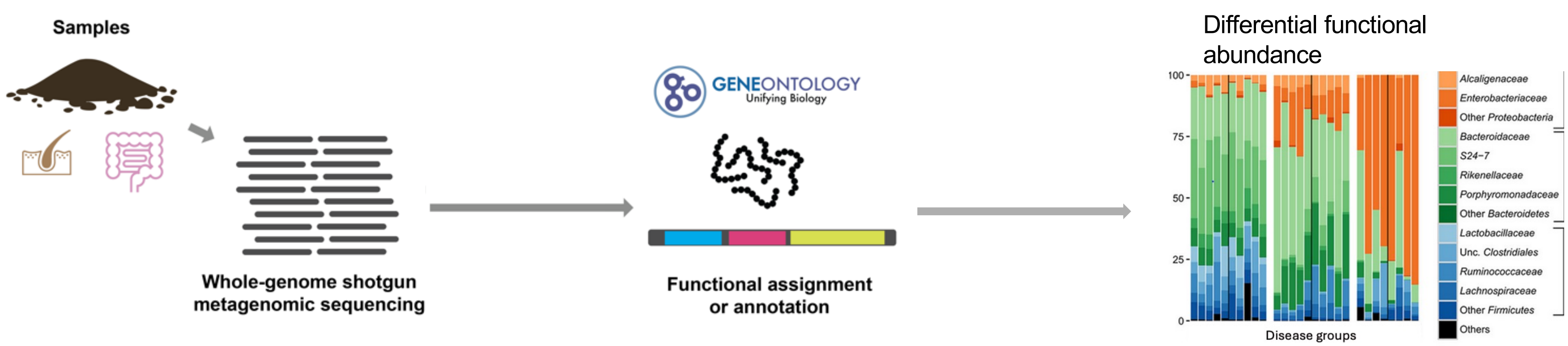
Future directions for microbiome research





Future directions for microbiome research





Open challenge in the field: It is difficult to figure out what bacteria do

There are a lot of bacteria and all of them have their own genomes

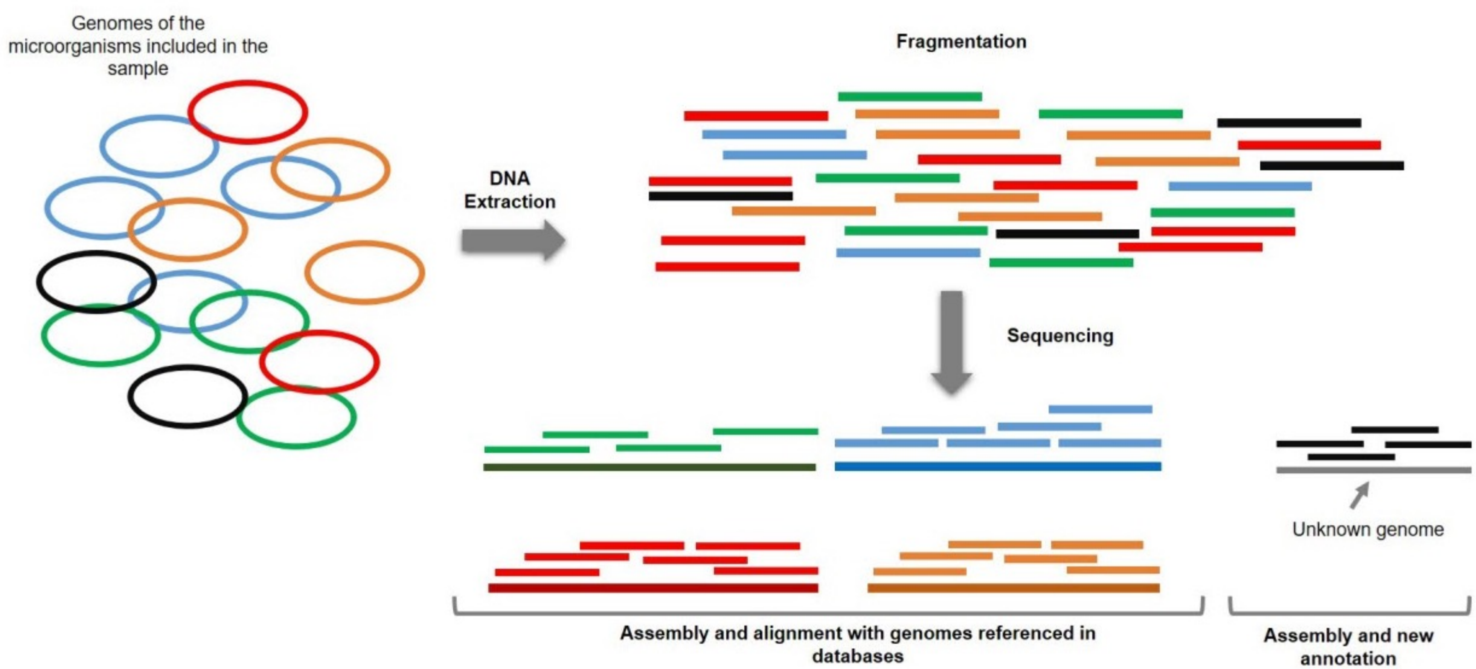


Image: France Genomique

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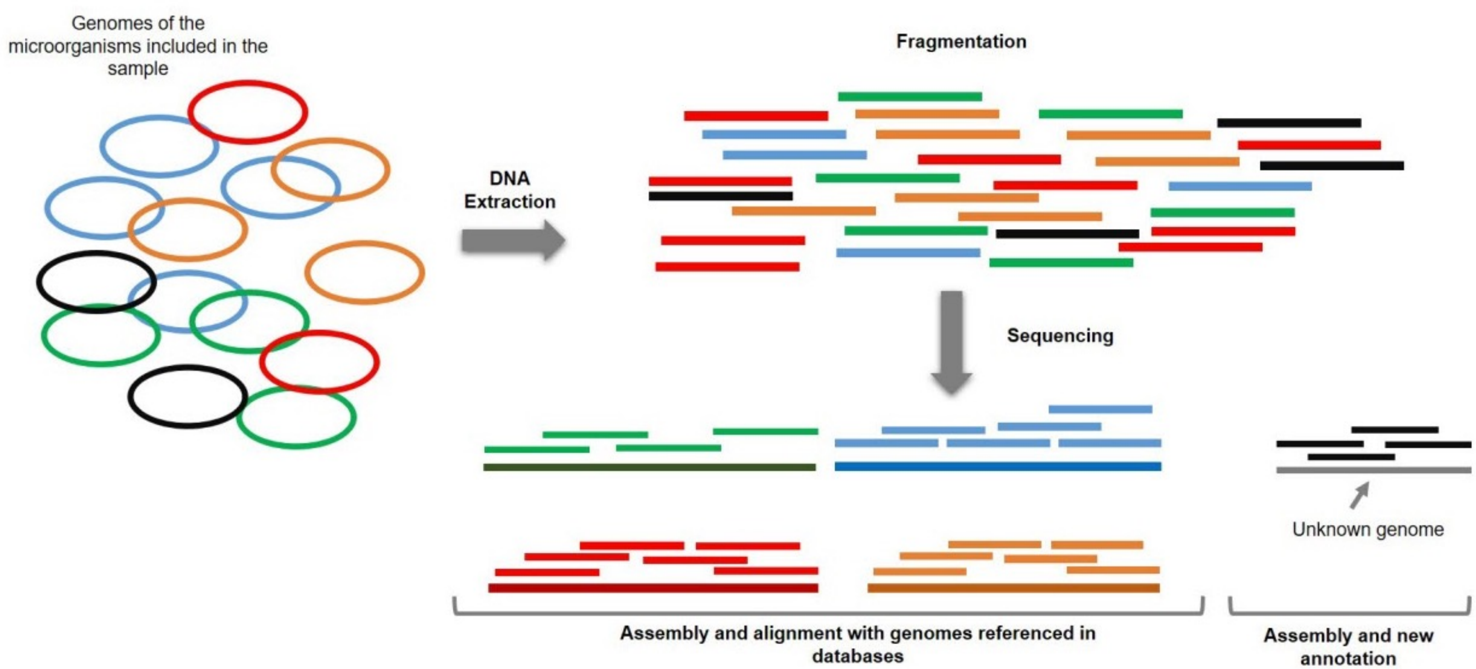
Many bacteria have genes with similar functions
These regions of the genome are highly similar in amino acid sequence



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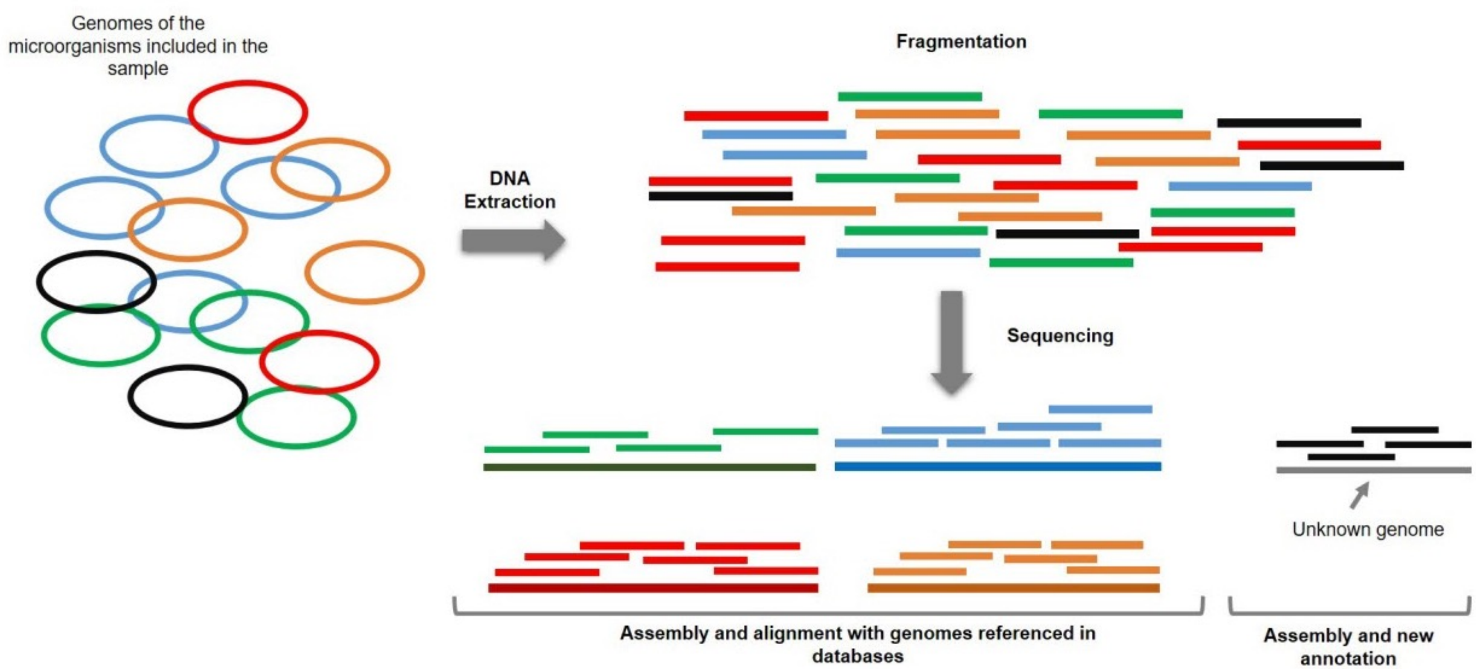


Gene sequences are like puzzle pieces
If two puzzles have the same piece, how can you tell which puzzle it goes to?

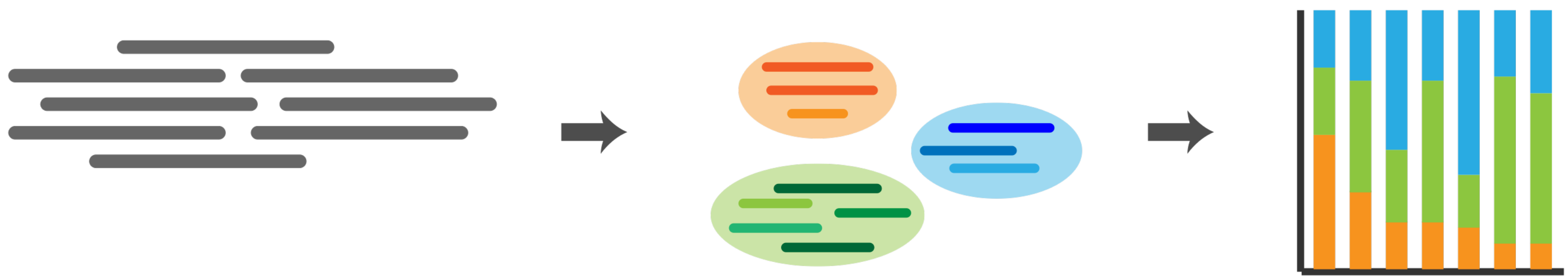
Image: France Genomique

We developed a method to quantify microbial gene products in a taxa-agnostic way

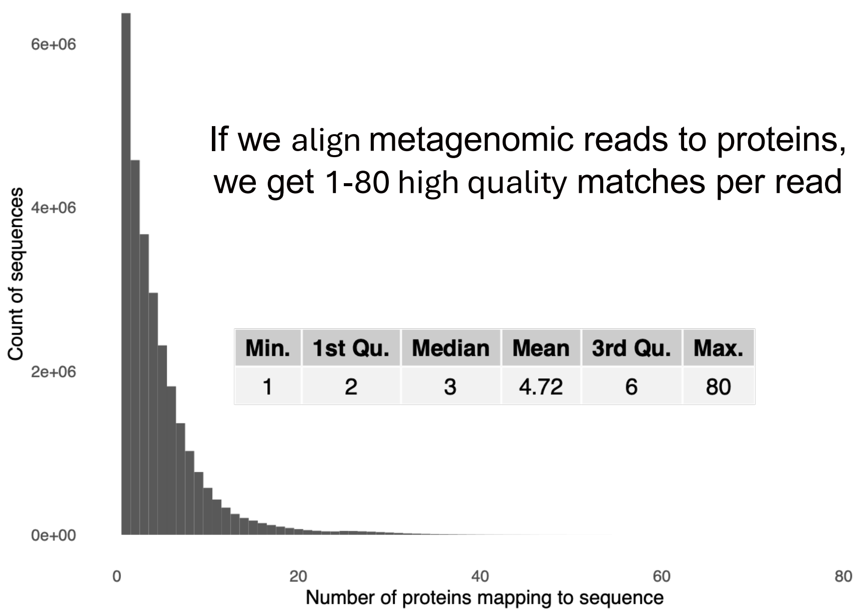
Whole-genome metagenomic shotgun sequencing

Read mapping to homologous taxa-agnostic protein clusters

Quantifcation at the protein cluster level

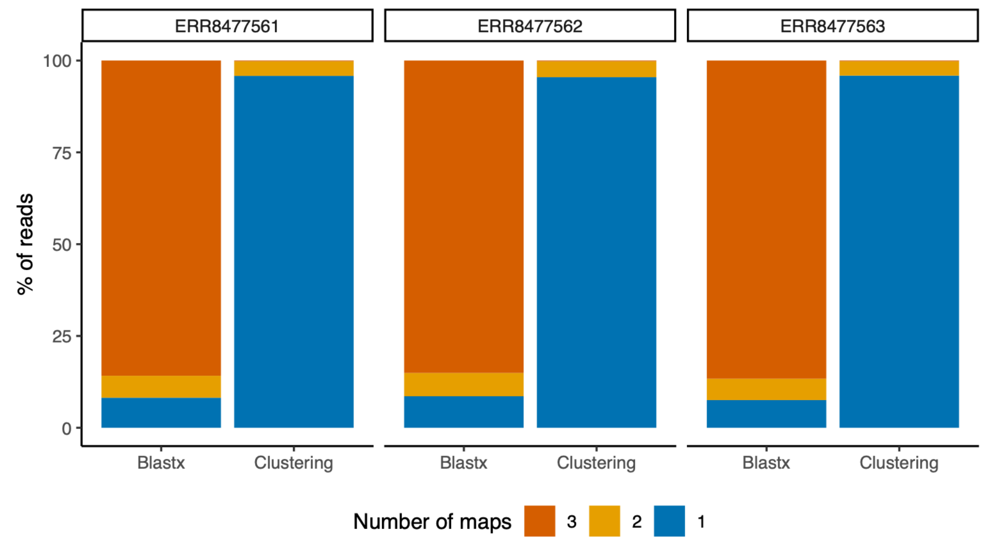


How does this method work?



If we align metagenomic reads to proteins, we get 1-80 high quality matches per read

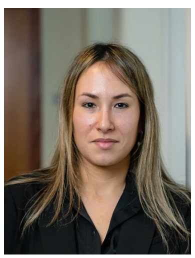
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	2	3	4.72	6	80



But the results are usually homologous proteins originating in different taxa

*k*Mermaid maps reads to taxa-agnostic protein groups using *k*-mers

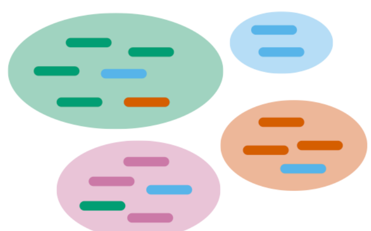
Daniel Schäffer



Dr. Noam Auslander

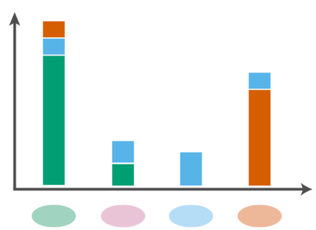
kMermaid

User inputs a nucleotide query sequence (top) and *k*Mermaid-provided *k*-mer frequency model (bottom)

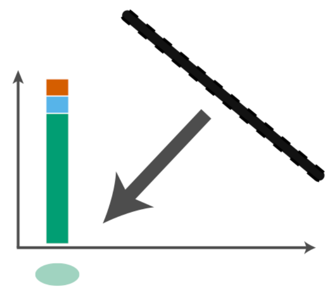


Step 1: A 6-frame translation is performed on the query sequence and frames containing one or more stop codons are discarded

Step 2: Open reading frames are fragmented into amino acid *k*-mers

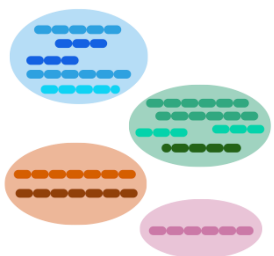


Step 3: Cluster *k*-mer frequencies are summed over each *k*-mer in the query sequence giving each cluster a composite mapping score

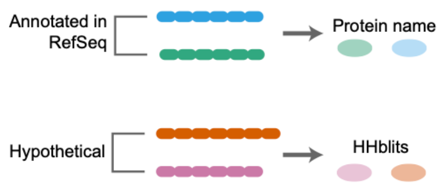


Step 4: The query sequence is assigned to the protein cluster with the highest composite frequency score

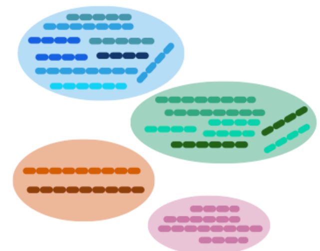
*k*Mermaid maps reads to taxa-agnostic protein groups using *k*-mers



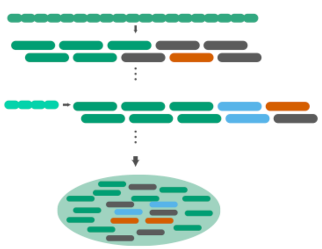
Step 1: Proteins from nr are greedily clustered by sequence similarity using CD-HIT at a lenient threshold



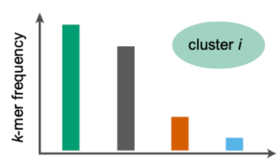
Step 2: Clusters are annotated by a representative sequence. If the cluster representative is a hypothetical protein, the closest hhblits annotation is used.



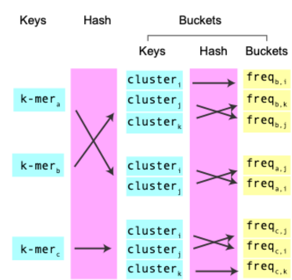
Step 3: All bacterial proteins in RefSeq are clustered into the established clusters using CD-HIT-2D



Step 4: All sequences in each cluster are fragmented into *k*-mers



Step 5: The per-cluster *k*-mer frequencies are computed for each *k*-mer in each cluster



Step 6: The cluster annotations and kmer frequencies are stored in a double hash map

$$s_C = \sum_{w \in C} frequency_{Cw_i}$$

s_c is the assignment score for read i to cluster C

$frequency_{Cw_i}$ is the frequency of *k*-mer i from the query read in cluster C

The read is assigned to the cluster with the max s_C

Objectives

- Define the microbiome
- Understand the microbiome’s contributions to human health and disease
- Know the two main experimental techniques for studying the microbiome
- Describe traditional microbiome metrics and analysis methods
  - + considerations for microbiome analysis
  - + review microbiome data resources
  - + new paradigm & function-first approaches
- **Questions?**